

High-Rate Communication for Underwater Acoustic Channels Using Multiple Transmitters and Space–Time Coding: Receiver Structures and Experimental Results

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Abstract—In this paper, we consider the use of multiple antennas and space–time coding for high data rate underwater acoustic (UWA) communications. Recent advances in information theory have shown that significant capacity gains can be achieved by using multiple-input–multiple-output (MIMO) systems and space–time coding techniques for rich scattering environments. This is especially significant for the UWA channel where the usable bandwidth is severely limited due to frequency-dependent attenuation. In this paper, we propose to use space–time coding and iterative decoding techniques to obtain high data rates and reliability over shallow-water, medium-range UWA channels. In particular, we propose to use space–time trellis codes (STTCs), layered space–time codes (LSTCs) and their combinations along with three low-complexity adaptive equalizer structures at the receiver. We consider multiband transmissions where the available bandwidth is divided into several subbands with guard bands in between them. We describe the theoretical basis of the proposed receivers along with a comprehensive set of experimental results obtained by processing data collected from real UWA communications experiments carried out in the Pacific Ocean. We demonstrate that by using space–time coding at the transmitter and sophisticated iterative processing at the receiver, we can obtain data rates and spectral efficiencies that are not possible with single transmitter systems at similar ranges and depths. In particular, we have demonstrated reliable transmission at a data rate of 48 kb/s in 23 kHz of bandwidth, and 12 kb/s in 3 kHz of bandwidth (a spectral efficiency of $4 \text{ b}\cdot\text{s}^{-1}\cdot\text{Hz}^{-1}$) at a 2-km range.

Index Terms—Iterative equalization, multiple-input–multiple-output (MIMO), space–time codes, successive interference cancellation (SIC), turbo codes, underwater acoustic (UWA) communications.

Manuscript received April 18, 2005; revised January 29, 2006; accepted February 22, 2007. This work was supported in part by the SPAWAR under In-house Laboratory Independent Research (ILIR) program and the U.S. Office of Naval Research (ONR) under SignalEx project for the construction of the MIMO transmit system. The 2005 Makai Experiment was primarily sponsored by the U.S. ONR under High Frequency Initiative.

Associate Editor: R. C. Spindel.

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Digital Object Identifier 10.1109/JOE.2007.899275

I. INTRODUCTION

RECENT information theoretic studies [1], [2] have shown that significant capacity improvement is possible with the use of multiple-input–multiple-output (MIMO) systems as opposed to more conventional single-input–single-output (SISO) systems. As an example, for a system employing n_T transmit and n_R receive antennas over a flat Rayleigh-fading channel, the information theoretic capacity grows linearly with the minimum of n_T and n_R . This tremendous increase in capacity, which directly translates into a corresponding increase in the achievable data rate, motivates us to investigate the performance of MIMO systems, for real underwater acoustic (UWA), frequency-selective-fading channels.

Typically, shallow-water, horizontal UWA channels are characterized by fast temporal variations, long multipath spreads that cause intersymbol interference (ISI), and most importantly, severe frequency-dependent attenuation. For instance, in long-range systems operating over several tens of kilometers, the available bandwidth is only on the order of several kilohertz. At medium ranges (several kilometers), the available channel bandwidth is on the order of several tens of kilohertz. Only at short ranges (a few tens of meters), the available bandwidth is in excess of 100 kHz [3], [4].

Given these bandwidth limitations for UWA channels, the data rates and spectral efficiencies achieved by existing UWA systems are typically much lower than what is achieved in radio channels. The earliest UWA modems (before 1990) used frequency-shift keying (FSK) which encodes information as discrete tones with guard times and guard bands inserted between the tones. Although these systems carefully avoided ISI, making them easy to implement, the data rates achieved are low (less than 1 kb/s) [5]. Only a decade ago it was demonstrated in [6] that bandwidth efficient phase coherent communications is possible for UWA channels by using receiver structures that jointly equalize and track the channel phase adaptively. These coherent schemes demonstrated significant improvements in data rate over noncoherent methods and achieved data rates of 10 kb/s in short-range, shallow-water UWA channels [6]. Further improvements in performance were obtained by successful application of receiver diversity in [7]. A comprehensive survey of the prevalent UWA systems can be found in [5].

However, to fundamentally improve the achieved data rates of the UWA channel over long transmission ranges, a cue must

be taken from the information theoretic works of [1] and [2] and the applicability of space-time coding over UWA channels must be investigated. The use of multiple antennas at the transmitter and the receiver provides us with additional degrees of freedom that can be used to obtain high spectral efficiencies and/or large diversity (particularly important for the bandlimited and highly dynamic UWA channel). It is also important to realize that, for UWA channels with large ISI spans, the optimal signal detection methods (maximum-likelihood sequence estimation (MLSE) [8] and maximum *a posteriori* probability (MAP) [9]) become prohibitively complex. For systems with multiple transmit antennas and large constellations, this complexity also grows exponentially with the number of transmit antennas and the constellation size (bits per symbol). This motivates us to develop novel low-complexity receiver structures that provide satisfactory performance.

Spatial modulation schemes using multiple transmitters and receivers for UWA channels are presented in [10]. The focus of [10] is to assess the number and nature of the available degrees of freedom in the UWA channel, rather than evaluating the performance of specific space-time codes. Using transmitter side channel state information (CSI), beamforming is performed and postequalization signal-to-noise ratio (SNR) results are reported along with *analytically inferred* bit error rates (BER) and channel capacities. Measured BER and actual achieved data rates are not presented.

In this paper, we propose to perform a comprehensive performance investigation of space-time coding techniques and novel receiver structures for real UWA channels, in terms of achieved data rates and measured BERs. We propose to use techniques such as space-time trellis codes (STTCs), layered space-time codes (LSTCs), and their combinations for high reliability (i.e., large diversity order) and high data rate communications over the UWA channel. To obtain a practical compromise between receiver complexity and carrier phase tracking capability, we consider the use of multiband transmissions where the available bandwidth is divided into several subbands with guard bands in between. This is a compromise between single carrier communication and orthogonal frequency-division multiplexing (OFDM). In the former, the symbol duration is small, and hence, the ISI span (measured in symbols) is large, requiring very long equalizers. However, small symbol durations enable better phase tracking since the channel remains coherent over a large number of symbols. In the latter, the symbol duration is larger than the multipath spread, and hence, ISI is mitigated, but Doppler estimation and channel tracking becomes a problem in this case since the coherence time of the channel is now much smaller in terms of the OFDM symbols. Thus, multiband communication, where a shorter equalizer can be used for each band, is an effective compromise between the two scenarios. We note that one potential drawback of multiband communication is its relatively high-peak-to-average-power ratio, which requires designing of power amplifiers with large, linear dynamic range.

Keeping the complexity issues in mind, we also propose several low-complexity adaptive MIMO equalizers in conjunction with suitable iterative decoding techniques. The proposed equalizers are predominantly based on the canonical decision feedback equalizer (DFE) [8], [11] structure with several

modifications made to tailor them to the UWA channel. These equalizers are also iterative in nature, which allows them to exchange probabilistic information with the channel decoder in a closed loop, thereby significantly reducing the error rates one would normally see for such high data rate transmissions. We describe the theoretical aspects of the proposed receiver structures and also present a comprehensive set of experimental results. The results are obtained by processing real at-sea data collected in a highly dynamic UWA channel in the Pacific Ocean. These results demonstrate that very high data rates and spectral efficiencies, which cannot be obtained by conventional single transmit antenna systems, can be obtained using the MIMO technology and space-time coding at similar ranges and depths.

This paper is organized as follows. Section II describes the MIMO transmission schemes we have investigated. Section III briefly describes the canonical MIMO DFE structure with embedded MIMO phase-locked loop (PLL) [12] which is an extension of the structure proposed in [6]. Section IV describes the MIMO DFE with embedded Viterbi decoding algorithm that uses STTCs without channel interleaving. Section V describes the MIMO DFE with iterative equalization. This MIMO iterative DFE is an extension of the SISO structure originally proposed in [13]. Section VI develops a successive-interference-cancellation-based receiver with adaptive channel estimation. This receiver, which has a serial structure, can also be used for iterative equalization following the principles of Section V. Section VII presents decoding results from a recent experiment for several different types of data sets ranging from single transmitter signals to six-transmitter MIMO signals for both binary phase-shift keying (BPSK) and quaternary phase-shift keying (QPSK) modulation schemes. Finally, Section VIII summarizes our results and discusses some interesting areas of future research.

II. TRANSMISSION SCHEMES

Consider a MIMO system with n_T transmit and n_R receive antennas. The signal transmitted from each transmit antenna can be encoded in several different ways to obtain either high spectral efficiency or large diversity (i.e., reliability) or, in some cases, a combination of both. In the following sections, we describe various schemes that we have investigated for MIMO UWA channels.

A. Space-Time Trellis Coding Without Symbol Interleaver

STTCs were introduced by Tarokh *et al.* in 1998 [14]. These codes are described by a trellis structure; an example is shown in Fig. 1. The incoming symbol stream is first encoded using the trellis structure and the encoded stream is then distributed among the transmit antennas. The trellis is designed to provide the full spatial diversity advantage $n_T n_R$. The example shown in Fig. 1 is designed for two transmit antennas using BPSK symbols, has four trellis states, and provides a transmit diversity of two. Similar codes exist for QPSK and 8-PSK modulation using a larger number of states. Decoding of these codes can be easily accomplished using the Viterbi algorithm operating on the trellis with a suitable distance metric.

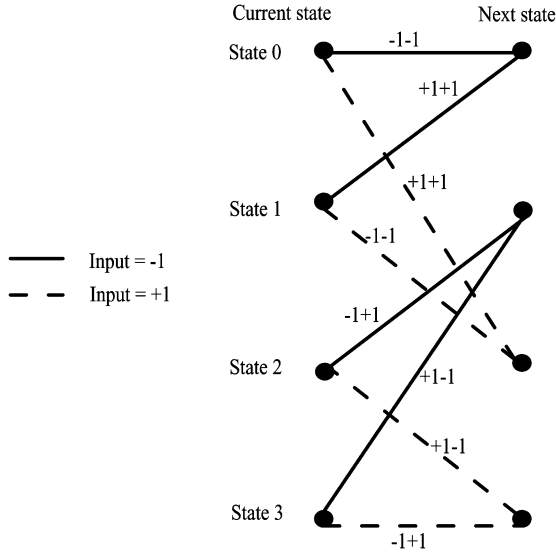
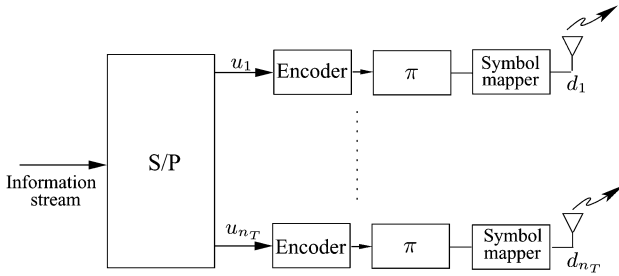

 Fig. 1. Trellis structure of four-state STTC, BPSK, $1 \text{ b} \cdot \text{s}^{-1} \cdot \text{Hz}^{-1}$.


Fig. 2. Layered space-time encoder.

B. Layered Space-Time Coding With Turbo Codes

LSTCs were introduced by Foschini *et al.* [15], [16], for rich scattering, flat Rayleigh-fading environments. Unlike STTC, which exploits the full spatial diversity of the system, these codes aim at achieving very high spectral efficiencies possible in a system with a large number of transmit antennas.

At the transmitter, the incoming bit stream is spatially multiplexed across the transmit antennas to form several substreams. Each substream is independently encoded using a turbo code, interleaved, and mapped into symbols before being transmitted over the channel, as shown in Fig. 2. Since independent streams are transmitted from each transmit antenna, the spectral efficiency of the system grows linearly with the number of transmit antennas. At the receiver, each receive antenna observes a superposition of all the streams corrupted by additive white Gaussian noise (AWGN). For flat-fading channels, these codes can be decoded using the suboptimal layered interference cancellation and nulling techniques [15]. Although these codes provide very high spectral efficiency, the diversity order obtained by them (using the technique of [15]) is only $n_R - n_T + 1$.

Turbo codes belong to the general class of capacity achieving, parallel concatenated convolutional codes where the encoder is formed by concatenating two recursive systematic convolutional codes (RSCCs) in parallel with a randomly generated interleaver, as shown in Fig. 3. The information bits are divided

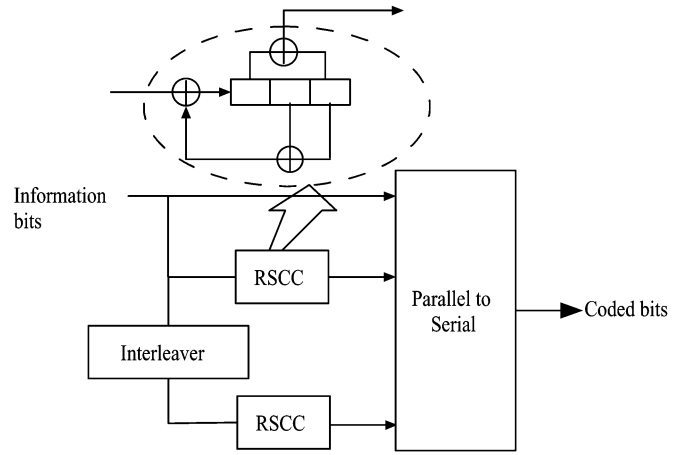


Fig. 3. Turbo encoder.

into blocks of length N . Each block is encoded using an RSCC, interleaved, and encoded using another RSCC (which may be identical to the first one). The systematic bits of the first and second encoder constitute the codeword corresponding to the information block. In this example, the code rate is $1/3$ since each information bit is mapped into three coded bits; however, higher code rates can also be obtained by puncturing mechanisms. At the receiver, the decoder employs a powerful iterative decoding algorithm to decode each information block by exchanging probabilistic information between the two constituent decoders (corresponding to the two RSCCs) [17]. The presence of the random interleaver within the turbo code provides a significant performance gain called *interleaver gain* which is one of the main reasons for near-capacity performance of turbo codes.

For our LSTC transmission schemes, we use a turbo code for each of the transmitters. The output of the encoder is further interleaved before being mapped onto the channel symbols to facilitate the iterative equalization process which we will describe in the following.

C. Grouped Space-Time Trellis Coding With Symbol Interleaver

To achieve high spectral efficiency along with reliability, we transmit several independent bit streams in parallel and encode each stream with an STTC, as shown in Fig. 4 [18]. At the receiver, we resort to powerful equalization algorithms that iteratively exchange probabilistic information between the equalizer and the space-time trellis decoder, thereby reducing the error rates significantly. It has been shown in [19] that for such iterative algorithms, an interleaver before transmission provides an additional interleaver gain. Therefore, in our experiments, we interleave each of the trellis-coded streams with a random interleaver before transmitting, as shown in Fig. 4. To ensure that the code properties are retained after interleaving, it is necessary to use the same interleaver for both streams of each STTC. Although we could use different interleavers for different groups (since the groups are independent of each other), we chose to use the same interleaver for all the groups for simplicity.

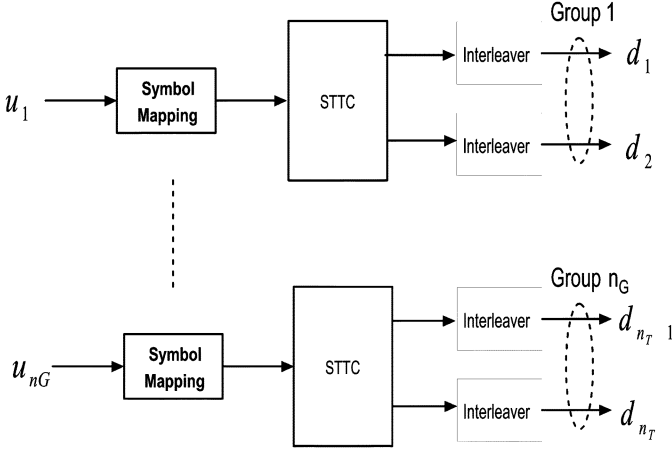


Fig. 4. Grouped space-time trellis encoding.

III. MIMO DFE WITH PLL

In this section, we briefly describe the canonical DFE-PLL structure that we have originally proposed in [12] as an extension of [6] for MIMO systems.

A. System Model

Consider a MIMO system with n_T transmit and n_R receive antennas. The signal transmitted from the i th transmit antenna can be represented in the baseband form as

$$u_i(t) = \sum_n d_i(n)g(t - nT), \quad 1 \leq i \leq n_T \quad (1)$$

where $d_i(n)$ is the space-time coded M -ary symbol transmitted from the i th transmit antenna at time instant n , $g(t)$ is the transmit pulse shape, which is assumed to be the same for all the transmit antennas, and T is the symbol duration. The continuous-time received signal at the j th receive antenna after matched filtering can be written as

$$v_j(t) = \sum_{i=1}^{n_T} \sum_n d_i(n)h_{ij}(t - nT - \tau_{ij}(t))e^{j\theta_{ij}(t)} + \nu_j(t) \quad (2)$$

where $h_{ij}(t)$ is the overall channel-impulse response between the i th transmit and the j th receive antenna, which includes both the transmit and the receive filters, as well as the physical channel. The variables $\theta_{ij}(t)$ and $\tau_{ij}(t)$ are the corresponding carrier phase distortion and the delay uncertainty, respectively, while $\nu_j(t)$ is the AWGN process at the j th receive antenna.

Equation (2) indicates that, for a MIMO system over a frequency-selective-fading channel, the transmitted symbols from each transmit antenna, not only experience ISI, but also cochannel interference (CCI) from the other transmit antennas. Thus, the receiver has to mitigate the effects of both ISI and CCI, while compensating for the rapid temporal variations of the UWA channel. Section III-B describes the MIMO-DFE structure, which is jointly optimized with a second-order PLL to perform equalization and synchronization.

B. Receiver Structure

The jointly optimized MIMO-DFE-PLL structure is shown in Fig. 5. The filter feedforward section consists of n_T filter banks, where the i th bank aims to equalize the i th data stream. Each bank consists of n_R finite-impulse response (FIR) filters, one for each of the n_R incoming received signal streams. The j th filter of the i th bank ($1 \leq i \leq n_T, 1 \leq j \leq n_R$) is followed by a phase compensator (derotator), which explicitly tracks the carrier phase variation caused by the channel between the i th transmitter and the j th receiver. While in theory an equalizer can compensate for phase variations introduced by the channel, this may not be the case in practice. Particularly for the UWA channel, the presence of residual carrier frequency offset causes equalizer tap rotation, ultimately causing the filters to diverge. Thus, the presence of an explicit phase compensator for each link of the MIMO system is essential for proper equalizer operation [6].

Each FIR filter in the feedforward (FF) section consists of a causal and a noncausal part to mitigate ISI due to the future and the past symbols, respectively. The output of the feedforward section for the data stream i is denoted as $p_i(n)$. The receiver feedback section consists of n_T filter banks. The aim of the i th filter bank is to mitigate the residual ISI and the CCI experienced by the i th symbol stream. The feedback filter output for the i th stream is denoted as $q_i(n)$. The overall DFE output for the i th stream is formed as

$$\hat{d}_i(n) = p_i(n) - q_i(n). \quad (3)$$

Maximum-likelihood detection is then performed on $\hat{d}_i(n)$ and a hard decision on the transmitted symbol $\tilde{d}_i(n)$ is made. The estimation error is defined as

$$e_i(n) = d_i(n) - \hat{d}_i(n) \quad (4)$$

where $d_i(n)$ is the known transmitted symbol during the training mode and is replaced by $\tilde{d}_i(n)$ during the decision directed mode. The parameters for equalization of the i th stream are obtained by minimizing the mean squared error (MSE) $\text{MSE}_i = E[|e_i(n)|^2]$.

The parameters to be jointly optimized are the equalizer coefficient vectors and the synchronization parameters, namely, the channel phase and timing estimates $\hat{\theta}_{ij}$ and $\hat{\tau}_{ij}$, respectively, ($1 \leq i \leq n_T, 1 \leq j \leq n_R$). To adaptively update the equalizer coefficients, the recursive least squares (RLS) algorithm [20] is used with $e_i(n)$ as the error sequence driving the equalizer. The channel phase and the symbol timing parameters are estimated using a second-order gradient-based algorithm. The gradients of the MSE, with respect to $\hat{\theta}_{ij}$ and $\hat{\tau}_{ij}$, are given by

$$\frac{\partial \text{MSE}}{\partial \hat{\theta}_{ij}} = -2\Im[E\{p_{ij}(n)(p_{ij}(n) + e_i(n))^*\}] \quad (5)$$

$$\frac{\partial \text{MSE}}{\partial \hat{\tau}_{ij}} = -2\Re[E\{\dot{p}_{ij}(n)e_i^*(n)\}] \quad (6)$$

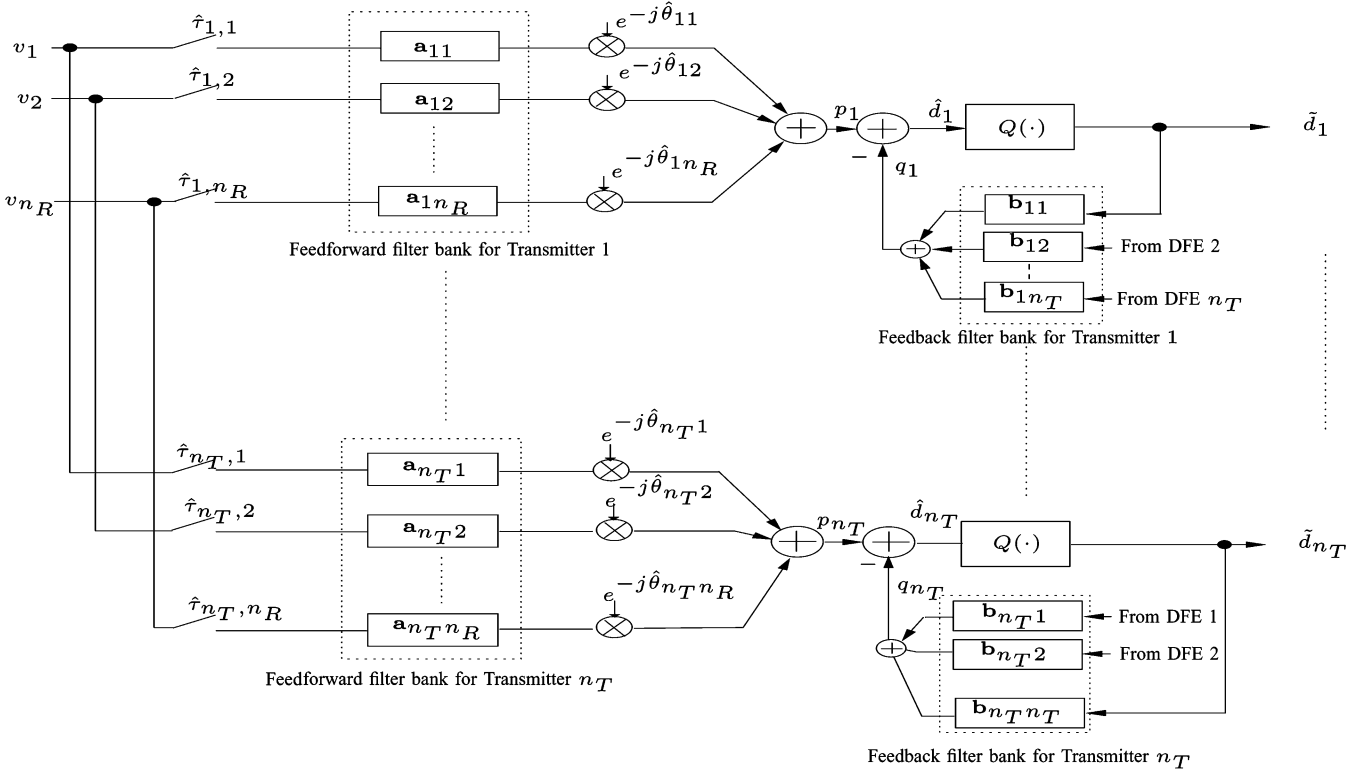


Fig. 5. MIMO DFE structure.

where $p_{ij}(n)$ is the output of the j th filter in the i th filter bank (for the i th stream), and $\dot{p}_{ij}(n)$ is the time derivative of $p_{ij}(n)$. The second-order phase update equation is given by

$$\hat{\theta}_{ij}(n+1) = \hat{\theta}_{ij}(n) + K_{f1}\Phi_{ij}(n) + K_{f2}\sum_{k=0}^n \Phi_{ij}(k) \quad (7)$$

where $\Phi_{ij}(n)$ is the instantaneous gradient of the MSE, with respect to $\hat{\theta}_{ij}$, i.e.,

$$\Phi_{ij}(n) = \Im[p_{ij}(n)(p_{ij}(n) + e_i(n))^*] \quad (8)$$

and K_{f1} and K_{f2} are the proportional and integral gain constants. A similar equation for the timing parameter can be obtained by forming the instantaneous gradient of the MSE with respect to $\hat{\tau}_{ij}$.

IV. MIMO DFE WITH EMBEDDED VITERBI FOR STTCs

The STTCs described in Section II-A provide full spatial diversity of $n_T n_R$ for flat Rayleigh-fading channels. To investigate the STTC performance over UWA channels, we propose a receiver structure for joint equalization and decoding of the STTC. The proposed structure consists of a DFE with an embedded space-time trellis decoder, implemented using the Viterbi algorithm, as shown in Fig. 6. However, the standard Viterbi algorithm produces symbol decisions with a certain delay, and hence, it is not suitable for use in the decision feedback process. Therefore, the Viterbi algorithm structure is modified to generate instantaneous (but tentative) symbol decisions, which are then fed back to the equalizer. Since these decisions are made using the powerful structure of the trellis code, they are much more reliable than the decisions made by a

symbol-by-symbol slicer, and thus, they help reduce DFE error propagation effects. In the rest of this section, we outline the Viterbi algorithm for the sake of completeness, and then explain the proposed joint equalization/decoding algorithm in detail.

Let $\hat{d}_i(n)$, $i = 1, 2, \dots, n_T$, denote the soft equalizer outputs at time instant n , and $d_i(s', s)$ denote the symbol transmitted from the i th transmit antenna, corresponding to the state transition $s' \rightarrow s$ of the trellis, at time n . The branch metric for each trellis branch is first computed as

$$\rho_n(s', s) = \sum_{i=1}^{n_T} \left| \hat{d}_i(n) - d_i(s', s) \right|^2. \quad (9)$$

The overall metric corresponding to each outgoing state at time n is then computed recursively as

$$\mu_n(s) = \min_{s'} [\rho_n(s', s) + \mu_{n-1}(s')] \quad (10)$$

where $\mu_k(s)$ denotes the overall metric accumulated at state s at time instant k . Thus, we have a candidate sequence (survivor) for each outgoing state, at every time instant, obtained by the aforementioned Viterbi algorithm. The hard decisions on the information symbols are then made by retracing the trellis from state 0 once the entire block is processed, thereby introducing a delay of one code block (in practice, processing the entire block may not be necessary and tentative hard decisions can be made with a smaller delay).

However, for a MIMO UWA channel, which is characterized by rapid temporal variations coupled with long ISI and severe CCI, this delay can be unacceptable. In other words, performing equalization and decoding separately can cause the DFE to suffer from excessive error propagation due to the unreliable decisions

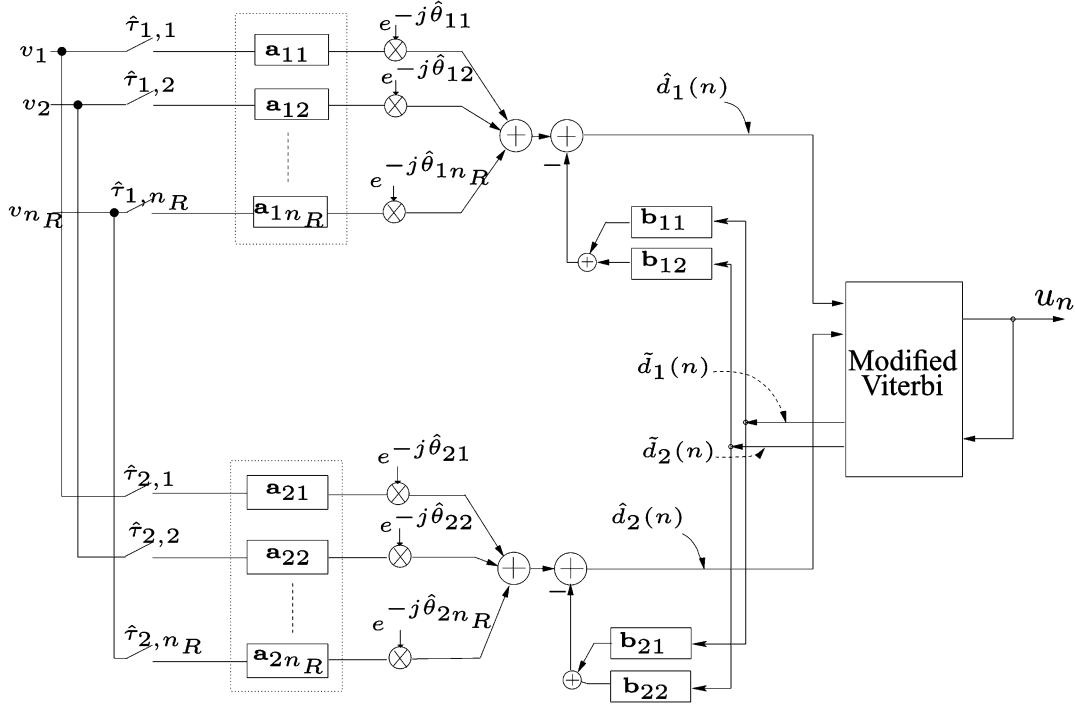


Fig. 6. MIMO DFE with embedded STTC decoder using modified Viterbi algorithm.

(obtained by a symbol-by-symbol slicer) being fed back to it during the equalization process. To overcome this problem, we propose a joint equalization/decoding strategy in which we feed back instantaneous but reliable symbol decisions to the equalizer by making use of the trellis structure of the STTC.

The proposed algorithm works as follows. At every time instant n , after the Viterbi algorithm computes the surviving branches, we compute the joint probability of each outgoing state s , and all the equalizer outputs from time 1 to time n . This is similar to what is done in the Bahl–Cocke–Jelinik–Raviv (BCJR) algorithm [9]. This joint probability is defined as

$$\alpha_n(s) \triangleq p(s, \mathbf{Y}_1^n) = p(s|\mathbf{Y}_1^n)p(\mathbf{Y}_1^n) \quad (11)$$

where $\mathbf{Y}_1^n = [\mathbf{y}_1, \dots, \mathbf{y}_n]$ and the vector $\mathbf{y}_i = [\hat{d}_1(i), \dots, \hat{d}_{n_T}(i)]^T$ is the vector at the equalizer output at time i . The variable $\alpha_n(s)$ is computed recursively as

$$\begin{aligned} \alpha_n(s) &= p(s, \mathbf{Y}_1^n) \\ &= \sum_{s'} p(s, s', \mathbf{Y}_1^n) \\ &= \sum_{s'} \gamma_n(s', s) \alpha_{n-1}(s') \end{aligned} \quad (12)$$

where $\gamma_n(s', s) = p(s, \mathbf{y}_n | s')$ is the state transition probability for the transition $s' \rightarrow s$. The variable $\gamma_n(s', s)$ is computed as

$$\begin{aligned} \gamma_n(s', s) &= p(s, \mathbf{y}_n | s') \\ &= p(\mathbf{y}_n | s, s') p(s | s') \\ &= A \exp \left[- \sum_{i=1}^{n_T} \frac{1}{2\sigma_i^2} \left| \hat{d}_i(n) - d_i(s', s) \right|^2 \right] p(u_n) \end{aligned} \quad (13)$$

where u_n is the *information* symbol corresponding to the transition $s' \rightarrow s$, σ_i^2 is the variance of the residual ISI and AWGN for the i th stream, and A is a constant. Assuming that all the symbols are equally likely [$p(u_n)$ does not depend on a specific state transition], we can rewrite (13) as

$$\gamma_n(s', s) = C_n \exp \left[- \sum_{i=1}^{n_T} \frac{1}{2\sigma_i^2} \left| \hat{d}_i(n) - d_i(s', s) \right|^2 \right] \quad (14)$$

where C_n is a constant, independent of s' or s . Using (12) and (14), the state probability α is computed for every outgoing state s at time n , and the state with the highest value of α is chosen for the instantaneous trellis traceback. The trellis is then read back from the chosen state at each time instant n and the coded symbols, so obtained, are fed back to the feedback portion of the equalizer. Clearly, by making use of the trellis structure in the equalization process, the tentative hard decisions for the DFE are much more reliable than the symbol-by-symbol decisions made by the memoryless slicer, thereby reducing the effects of error propagation considerably.

V. ITERATIVE MIMO DFE

It has been shown in [19] that for a coded system over frequency selective channels, significant performance improvement can be obtained by performing iterative (turbo) equalization where the equalizer and the channel decoder exchange soft information in an iterative fashion, thereby reducing the BER of the system gradually. Several low-complexity iterative structures for the UWA channel have been proposed in [4], [13], and [21]. In the rest of this section, we outline the extension of the iterative DFE structure proposed in [13] for MIMO systems.

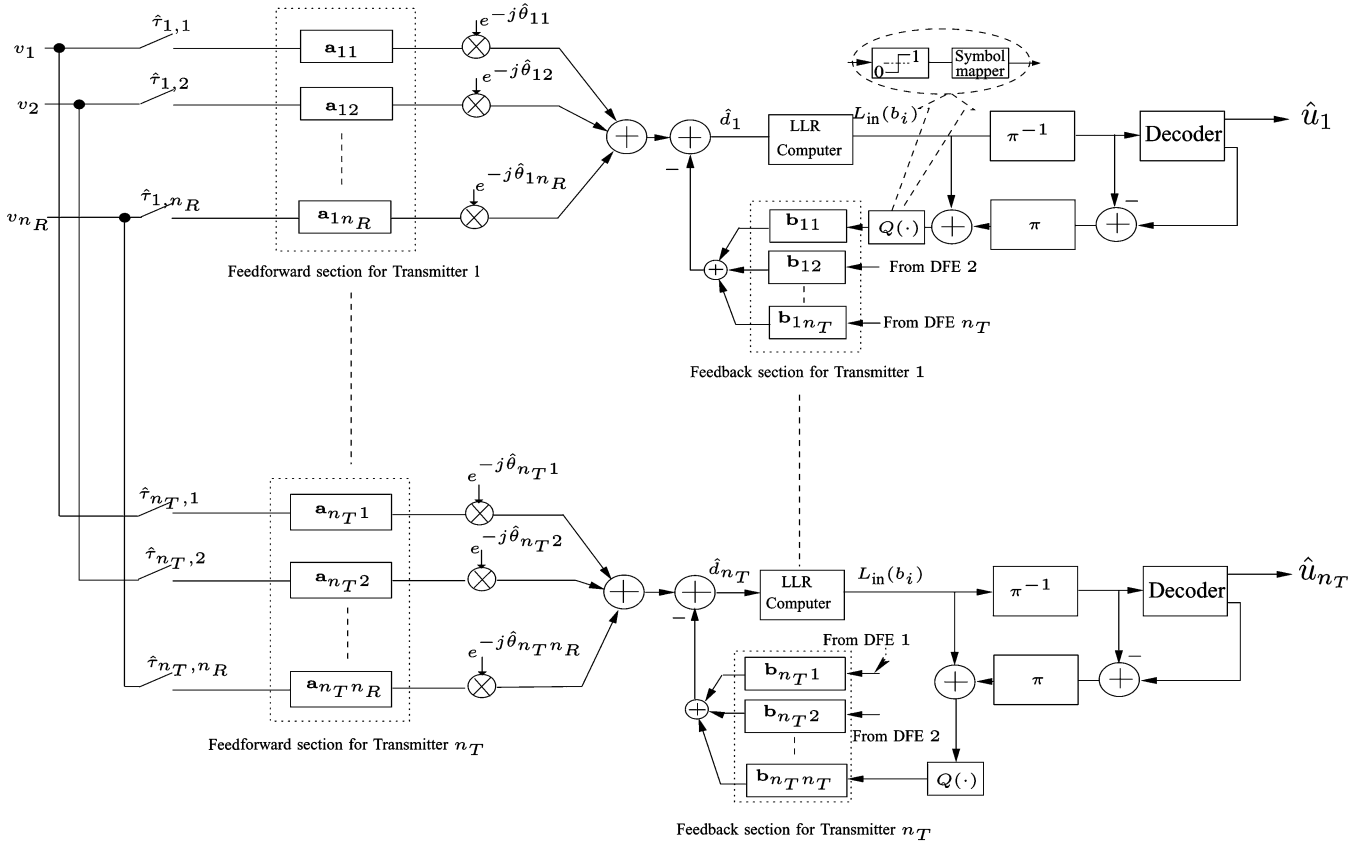


Fig. 7. Iterative MIMO DFE structure.

The iterative MIMO DFE block diagram is shown in Fig. 7. The equalizer output ($\hat{d}_i$) for each transmit antenna contains the soft estimate of the channel symbols. These estimates are used to compute the soft (coded) bit estimates of that symbol in the form of log-likelihood ratios (LLR). This is done as follows. An M -ary symbol is defined as $d = f(b_1, b_2, \dots, b_{\log_2 M})$ where b_i is the i th bit of the symbol and $f(\cdot)$ is the mapping from bits to symbols. If $\hat{d}$ is the soft symbol estimate of d , then the LLR for the i th bit of d is obtained as

$$\begin{aligned} L_{in}(b_i) &= \log \frac{\Pr(\hat{d}|b_i = 1)}{\Pr(\hat{d}|b_i = 0)} \\ &= \log \frac{\sum_{\forall d: b_i = 1} \Pr(\hat{d}|d)}{\sum_{\forall d: b_i = 0} \Pr(\hat{d}|d)} \\ &= \log \frac{\sum_{\forall d: b_i = 1} \exp(-|\hat{d} - d|^2 / 2\sigma^2)}{\sum_{\forall d: b_i = 0} \exp(-|\hat{d} - d|^2 / 2\sigma^2)}. \end{aligned} \quad (15)$$

In the previous equations, we have assumed that the equalizer output is Gaussian distributed with conditional variance σ^2 . This variance can be easily estimated from the error sequence at the equalizer output. The LLRs are then deinterleaved and passed on to the outer decoder which computes an estimate of the information symbols and also the overall LLR of the coded bits. The outer decoder must be a soft-input-soft-output module

which can be a MAP decoder [9] for convolutional codes or a turbo decoder if a turbo code is used. The input to the decoder is then subtracted from the output LLR to form the extrinsic (new) information about the coded bits which after suitable reinterleaving is passed to the DFE for the next turbo iteration. The DFE uses this new information in the next turbo iteration by adding it to the LLR of the coded bits obtained from the symbol estimates in that iteration. If we denote the extrinsic information for the i th bit available to the DFE as $L_{ext}(b_i)$, then the hard bit decisions fed back to the DFE feedback section are computed using the decision variable

$$L_{overall}(b_i) = L_{in}(b_i) + L_{ext}(b_i). \quad (16)$$

Thus, $\hat{b}_i = 1$ if $L_{overall}(b_i) \geq 0$ and $\hat{b}_i = 0$ if $L_{overall}(b_i) < 0$. The hard bit estimates are then grouped to form the hard symbol estimates which are then fed back to the DFE.

VI. ITERATIVE DFE WITH CHANNEL ESTIMATION AND SUCCESSIVE INTERFERENCE CANCELLATION (SIC) DFE

The discrete time equivalent of the signal received at the j th receive antenna (2) can be expressed by a tapped delay line model as

$$v_j(n) = \sum_{k=1}^{n_T} \left[\sum_{l=0}^D h_{kj}^{(l)}(n) d_k(n-l) \right] e^{j\theta_{kj}(n)} + \eta_j(n) \quad (17)$$

where $h_{kj}^{(l)}(n)$ is the l th tap of the $D+1$ tap impulse response vector from the k th transmitter to the j th receiver at time n .

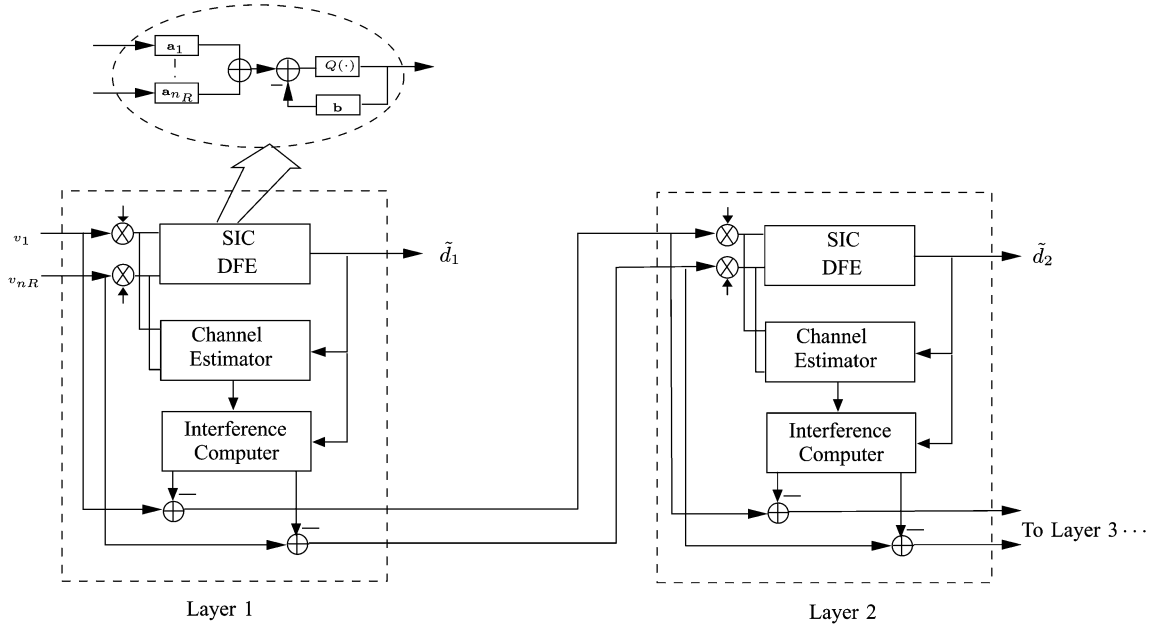


Fig. 8. DFE with SIC.

$h_{kj}^{(l)}(n)$ captures the small scale time variations of the channel whereas the phase rotation due to Doppler shift is treated separately using θ_{kj} [6]. If we define the column vectors

$$\mathbf{h}_{kj}(n) = [h_{kj}^{(0)}(n), h_{kj}^{(1)}(n), \dots, h_{kj}^{(D)}(n)]^T \quad (18)$$

and

$$\mathbf{D}_k(n) = [d_k(n), d_k(n-1), \dots, d_k(n-D)]^T \quad (19)$$

then (17) can be compactly written as

$$v_j(n) = \sum_{k=1}^{n_T} \mathbf{h}_{kj}^T(n) \mathbf{D}_k(n) e^{j\theta_{kj}(n)} + \eta_j(n) \quad 1 \leq j \leq n_R. \quad (20)$$

The received signal is phase-compensated for each transmitter using the corresponding PLL output. The phase-compensated received signal for the i th transmitter can be expressed as

$$\begin{aligned} r_{ij}(n) &= v_j(n) e^{-j\tilde{\theta}_{ij}(n)} \\ &\approx \mathbf{h}_{ij}^T(n) \mathbf{D}_i(n) + \sum_{k=1, k \neq i}^{n_T} \mathbf{h}_{kj}^T(n) \mathbf{D}_k(n) + \eta'_j(n) \end{aligned} \quad (21)$$

where $\tilde{\theta}_{ij}(n)$ is the channel phase estimate from the i th transmitter to the j th receiver.

At this point, our aim for the i th transmitter is twofold: 1) equalize the received signal and obtain the data symbol estimates $\tilde{d}_i(n)$, $\forall n$ and 2) estimate the channel-impulse response vector $\mathbf{h}_{ij}(n)$, $\forall n$. Of course, we also have the phase estimates $\tilde{\theta}_{ij}(n)$ from the corresponding PLL. Once we have the previous

quantities available for the i th transmitter we proceed to reconstruct the interference caused by that transmitter using (20) as

$$\tilde{I}_{ij}(n) = \tilde{\mathbf{h}}_{ij}^T(n) \tilde{\mathbf{D}}_i(n) e^{j\tilde{\theta}_{ij}(n)} \quad \forall n \quad (22)$$

and cancel its contribution from $v_j(n)$. In other words, we update

$$v_j(n) = v_j(n) - \tilde{I}_{ij}(n) \quad \forall n. \quad (23)$$

and use the updated received signal for the next transmitter, as shown in Fig. 8. The aforementioned procedure is repeated for all transmitters in a serial fashion until all the transmitters are equalized.

A. Equalization Using SIC DFE

The operation of the iterative SIC DFE is similar to that of the iterative MIMO DFE described in Section V. However, unlike the MIMO DFE, the feedback portion of each transmitter in an SIC DFE has feedback from the symbols of only that transmitter, as shown in Fig. 8. Since the feedback section for one transmitter does not require feedback from the other transmitters, this scheme allows us to detect the data symbols serially, in a layer-by-layer fashion and then perform interference cancellation serially [11]. Moreover, the feedback filter for each transmitter in this scheme is considerably shorter since it attempts to cancel only the residual ISI for that transmitter (not the CCI), leading to a considerable complexity reduction in the equalizer update process. Moreover, the iterative equalization technique described in Section V also applies to this serial structure allowing us to perform turbo equalization along with interference cancellation.

B. Estimation of the Impulse Response Vector $\mathbf{h}_{ij}(n)$

The phase-compensated signal $r_{ij}(n)$ in (21) is used as the input to the channel estimator. The estimator input for the i th transmitter and the j th receiver is formed as

$$\mathbf{v}_{ij}(n) = \begin{bmatrix} r_{ij}(n + N_1) \\ \vdots \\ r_{ij}(n + 1) \\ r_{ij}(n) \end{bmatrix}. \quad (24)$$

We now define the input-output correlation vector as

$$\begin{aligned} \mathbf{R}_{ij}(n) &= E[\mathbf{v}_{ij}(n)d_i^*(n)] \\ &= \begin{bmatrix} E[r_{ij}(n + N_1)d_i^*(n)] \\ \vdots \\ E[r_{ij}(n + 1)d_i^*(n)] \\ E[r_{ij}(n)d_i^*(n)] \end{bmatrix}. \end{aligned} \quad (25)$$

At this point, we assume that the data symbols are uncorrelated both in space and time¹ and have unit energy, i.e.,

$$E[d_i(n)d_j^*(m)] = \begin{cases} 1, & i = j \text{ and } n = m \\ 0, & \text{else} \end{cases}. \quad (26)$$

Substituting (21) in each of the terms in (25), we have

$$\begin{aligned} E[r_{ij}(n + m)d_i^*(n)] &= \mathbf{h}_{ij}^T(n + m)E[\mathbf{D}_i(n + m)d_i^*(n)] \\ &= h_{ij}^{(m)}(n + m), \quad 0 \leq m \leq N_1. \end{aligned} \quad (27)$$

Thus, each element of $\mathbf{R}_{ij}(n)$ in (25) contains an estimate of the complex channel coefficient, staggered in time. However, if we assume that the length of the ISI is much smaller compared to the coherence time of the channel, then we can approximate

$$h_{ij}^{(m)}(n + m) \approx h_{ij}^{(m)}(n), \quad 0 \leq m \leq N_1 \quad (28)$$

which is equivalent to assuming that the channel coefficients do not vary significantly within the estimator span. With this assumption, we see that

$$\mathbf{R}_{ij}(n) = \begin{bmatrix} h_{ij}^{(N_1)}(n) \\ \vdots \\ h_{ij}^{(1)}(n) \\ h_{ij}^{(0)}(n) \end{bmatrix}. \quad (29)$$

Thus, the input-output correlation vector is actually a time-reversed version of the desired impulse response. However, since the impulse response is time varying, we prefer to estimate $\mathbf{R}_{ij}(n)$ recursively. We do so by defining

$$\hat{\mathbf{R}}_{ij}(n) = \mu_n \sum_{k=1}^n \lambda_c^{n-k} \mathbf{v}_{ij}(k)d_i^*(k) \quad (30)$$

¹Since the data symbols are the components of a space-time codeword, the previously described assumption is not true in general. However, the presence of a channel interleaver sufficiently decorrelates the data sequence for practical purposes.

where λ_c is the forgetting factor for the channel estimator and μ_n is a monotonically decreasing function of n which we will optimize. As an example, if $\lambda_c = 1$, then one can choose $\mu_n = 1/n$ to obtain a linear unbiased time-averaged estimate of $\mathbf{R}_{ij}(n)$ in (25). In general, μ_n can be obtained by taking the expectation on both sides of (30). Assuming stationarity along with the requirement that $\hat{\mathbf{R}}_{ij}(n)$ should be an unbiased estimate of $\mathbf{R}_{ij}(n)$, we find that

$$\mu_n \sum_{k=1}^n \lambda_c^{n-k} = 1 \quad (31)$$

or

$$\mu_n = \frac{1 - \lambda_c}{1 - \lambda_c^n}. \quad (32)$$

The recursion for $\hat{\mathbf{R}}_{ij}(n)$ is then formulated as

$$\begin{aligned} \hat{\mathbf{R}}_{ij}(n) &= \mu_n \left[\lambda_c \sum_{k=1}^{n-1} \lambda_c^{n-1-k} \mathbf{v}_{ij}(k)d_i^*(k) \right] + \mu_n \mathbf{v}_{ij}(n)d_i^*(n) \\ &= \lambda_c \frac{\mu_n}{\mu_{n-1}} \hat{\mathbf{R}}_{ij}(n-1) + \mu_n \mathbf{v}_{ij}(n)d_i^*(n) \\ &= \frac{\lambda_c - \lambda_c^n}{1 - \lambda_c^n} \hat{\mathbf{R}}_{ij}(n-1) + \frac{1 - \lambda_c}{1 - \lambda_c^n} \mathbf{v}_{ij}(n)d_i^*(n) \end{aligned} \quad (33)$$

where d_i is the known symbol during the training mode and is replaced by the detected symbol $\hat{d}_i$ during the decision directed-mode. Thus, at every time instant, we have a time-reversed estimate of the impulse response vector between the i th transmitter and the j th receiver from the aforementioned recursion that can be used in the interference cancellation process. For practical systems, the feedforward section of the equalizer is usually fractionally spaced to make the structure robust to symbol timing errors. In that case, the symbol-spaced interference estimate obtained from (22) needs to be up-sampled before the interference cancellation procedure.

It is interesting to note that the aforementioned procedure for channel estimation can be viewed as an approximation to the RLS channel estimation algorithm where the problem is formulated as a system identification problem [4]. The system identification block is driven by the channel symbols and its output is compared with the channel output. In that case, the optimal algorithm in the least squares (LS) sense involves updating the input covariance matrix. However, since the input consists of independent and identically distributed symbols with equal energy, its covariance matrix will eventually converge to the identity matrix. Hence, to reduce complexity (especially for MIMO systems where a separate channel estimation block needs to be implemented for each link of the system), we force the input covariance matrix to identity from the very beginning, in which case the LS solution becomes the same as the one proposed previously.

VII. EXPERIMENTAL RESULTS

In this section, we provide a comprehensive set of experimental results based on decoding of real data transmitted through a UWA channel. We provide results for space-time coded signals using one, two, four, and six transmitters for both BPSK and QPSK modulation schemes.

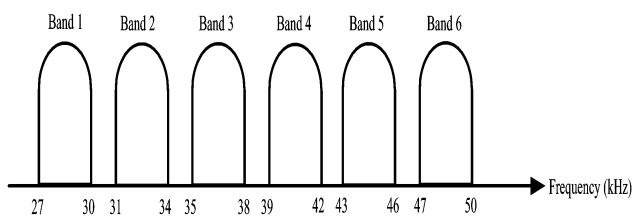


Fig. 9. Spectrum of the transmitted multiband signal.

A. Experiment Setup

A MIMO experiment named MakaiEx was conducted off the northwestern coast of Kauai, HI, in September 2005. Several space-time coding techniques such as STTC, grouped STTC, and turbo-coded LSTC, were investigated with both single and multiple carrier modulations. The transmissions were organized in packet structures to perform periodic channel estimation and synchronization, as shown in Fig. 11. Each packet consisted of a linear frequency-modulated (LFM) probe signal for both synchronization and channel estimation purposes, followed by a clearing time, and finally, followed by the transmission symbols. Training symbols were inserted into the packet to periodically retrain the equalizer.

Our transmissions consisted of both single and multiple carrier modulation. The single carrier data packets had a center frequency of 30 kHz. The symbol rate was 2000 symbols/s. The packet size was 4.8 s which resulted in 9600 symbols per packet. A square-root-raised cosine pulse with 50% excess bandwidth was used for pulse shaping, resulting in the overall bandwidth of 3 kHz. The multiband scheme had six carriers located at 28.5, 32.5, 36.5, 40.5, 44.5, and 48.5 kHz, as shown in Fig. 9. Each band had a symbol rate of 2000 symbols/s with 50% excess bandwidth, again resulting in 3-kHz bandwidth for each band. A guard band of 1 kHz was chosen between bands. Thus, the total bandwidth used was 23 kHz (from 27 to 50 kHz).

The transmitter consisted of a ten-element array with 2-m separation between elements and was deployed over the sides of the Kilo Moana research vessel. At nominal sound speeds and 30-kHz center frequency, the 2-m spacing corresponds to about 40 acoustic wavelengths between elements which is sufficient to ensure statistically independent channels for each transmitter-receiver pair. The receiver consisted of eight elements with a 2-m separation between the phones, and was allowed to drift freely while the Kilo Moana maintained a 2-km range from it. The source power level was set at 190 dB re 1 μ Pa @ 1 m.

Fig. 10 shows a sound-speed profile measured before the MIMO deployment during the Makai experiment, when the data set we used to obtain all of the results discussed in this paper was recorded. Our source array was deployed at depths spanning 40–67 m. Our receive array was deployed at depths spanning 20–44 m. Thus, both arrays are above the strong duct that starts at a depth of 70 m (although the deepest transmit element at 67 m may be somewhat affected by it, if the array is not kiting due to currents).

The space-time coding techniques tested mainly consisted of LSTC, STTC, and their combinations. The LSTC scheme used independent encoding on each transmitter using a rate 1/3 turbo code with component convolutional codes having generator polynomials $[1, (15/17)]_{\text{octal}}$. The interleaver of the code

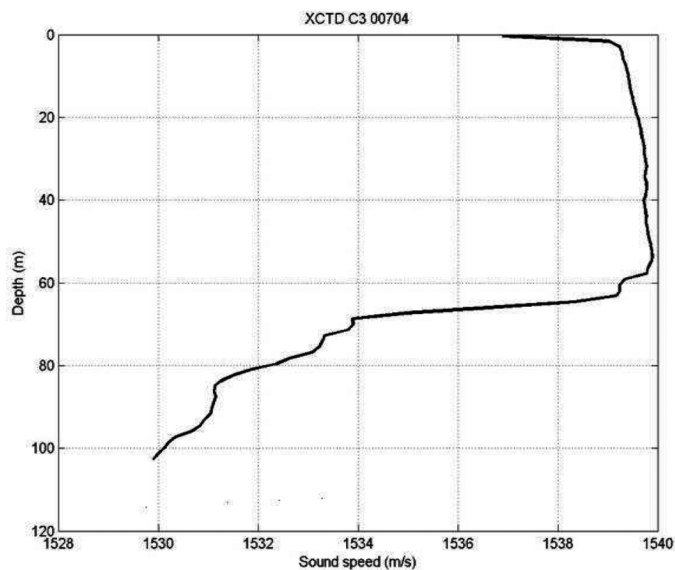


Fig. 10. Sound-speed profile during the experiment.

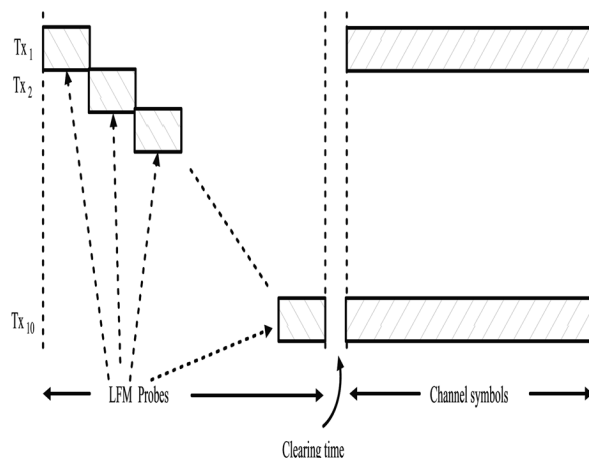


Fig. 11. Transmitted MIMO packet structure (two-transmitter example).

was random, and was chosen to be of length 3200. For the single antenna data packets, we used a rate 1/2 turbo code with the same generator polynomials but with an interleaver of size 4800.

The STTC schemes used the two-transmitter, four-state BPSK, four-state QPSK, and eight-state 8-PSK code of [14] followed by a symbol interleaver of size 9600 for each transmit antenna. The grouped STTC schemes consisted of 1, 2, and 3 groups of two-transmitter STTC transmitting simultaneously (i.e., using two, four, and six transmitters simultaneously).

B. Decoding Results

At the receiver, the first step in starting the decoding process is to synchronize (acquire) the packet. This is achieved by correlating the received packet with a local replica of the LFM probe. The peaks at the output of the correlator are used to determine the location of the packet. Moreover, since LFM probes have good correlation properties, the correlator output contains a time-averaged channel-impulse response estimate (Fig. 12), that can be used for setting the equalizer parameters later on.

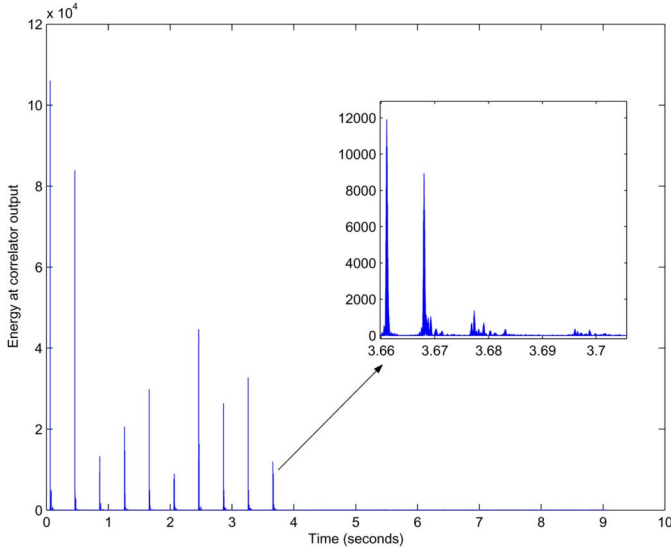


Fig. 12. Energy at the output of the correlator.

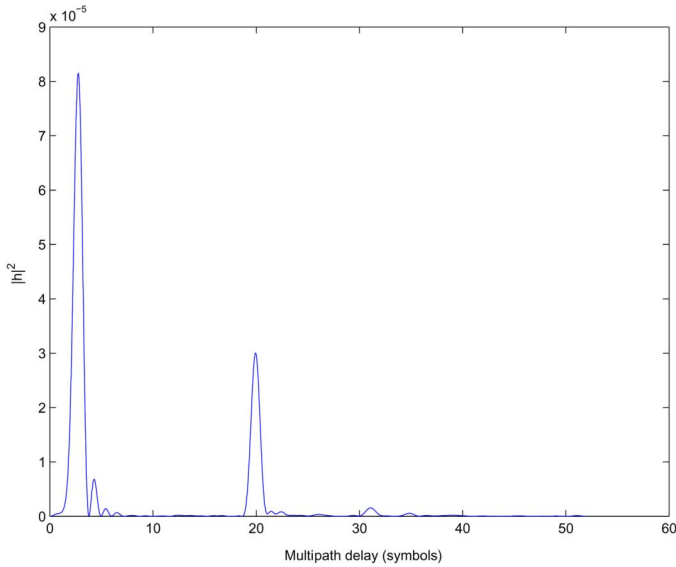


Fig. 13. Estimate of the channel-impulse response obtained using the RLS algorithm (Section VI) for the SISO data set.

Before processing each data packet, the desired equalizer parameters were chosen by examining the correlator output which contains a time-averaged channel-impulse response estimate for each transmitter–receiver pair of the system. The length of the impulse response allows us to choose the equalizer lengths. Next, to set the values for the RLS forgetting factor λ , and the PLL tracking constants, we decode a small block of known training symbols using the SIC DFE (with channel estimation). The channel-impulse response estimates at the DFE output provides a basis to set the desired quantities. For most of the packets processed so far, $\lambda = 0.999$ and tracking constants $K_{f1} = 0.01$ and $K_{f2} = 0.001$ have been found to be a good choice. To choose a subset of the eight available receivers, we compute the total impulse response energy observed at each receiver. If $\hat{\mathbf{h}}_{ij}(n)$ is the estimate of the instantaneous impulse

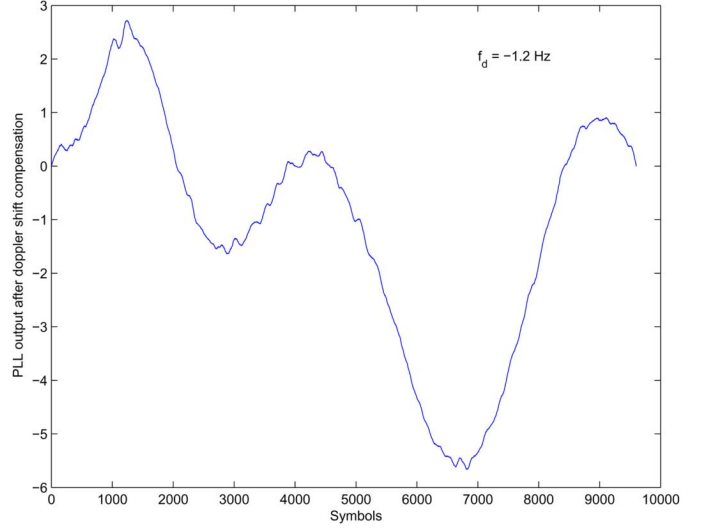


Fig. 14. Phase estimate at the PLL output after Doppler shift compensation for SISO data set.

TABLE I
DECODING RESULT FOR TURBO-CODED SISO DATA PACKET.
(n_T, n_R) = (1, 1). DATA RATE: 1 kb/s PER BAND, 6 kb/s TOTAL

Band	Symbol errors / 9600 (equalization only)	Information bit errors / 4800	SNR (dB)
1	651	0	7.18
2	7	0	12.24
3	4	0	14.60
4	9	0	12.12
5	57	0	8.92
6	280	0	6.26

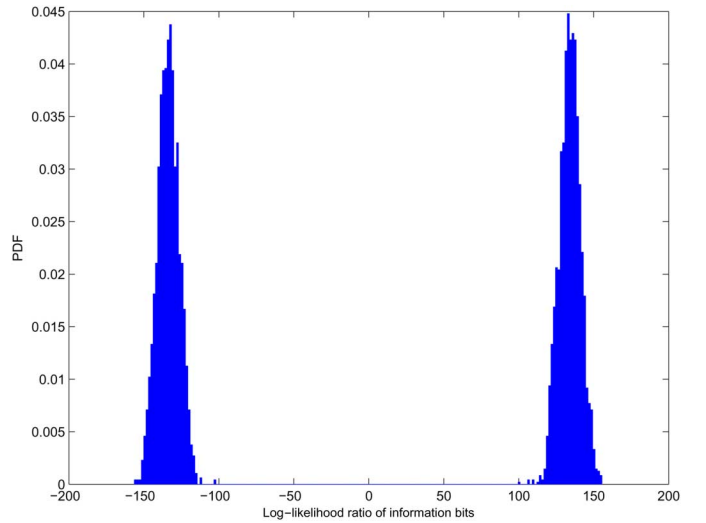


Fig. 15. Distribution of information bit LLRs for SISO data set.

response from transmitter i to receiver j , then the total energy at receiver j is computed as

$$\Upsilon_j = \sum_{i=1}^{n_T} \left\{ \frac{1}{T_i} \sum_{n=1}^{T_i} \|\hat{\mathbf{h}}_{ij}(n)\|^2 \right\} \quad (34)$$

where T_i is the length of the initial training block in symbols. The receiver subset with the highest energy is then chosen for subsequent decoding of the packet and is not changed within

TABLE II
DECODING RESULTS FOR SISO PACKETS OF TRANSMITTER 1 USING OPTIMIZED RECEIVER SETTING AND WITHOUT MANUAL INTERVENTION. "0" INDICATES SUCCESSFUL DECODING (ZERO ERRORS). "F" INDICATES DECODER FAILURE (CLOSE TO 50% ERRORS)

	Receiver 1	Receiver 2	Receiver 3	Receiver 4	Receiver 5	Receiver 6	Receiver 7	Receiver 8
Band 1	0	0	0	0	0	0	0	0
Band 2	0	0	0	0	0	0	0	0
Band 3	0	0	0	0	F	0	F	0
Band 4	0	F	0	F	0	0	0	0
Band 5	0	0	0	0	0	0	0	0
Band 6	0	F	0	F	0	0	F	F

TABLE III
DECODING RESULT FOR BPSK, TURBO-CODED, LSTC USING MIMO DFE. $(n_T, n_R) = (4, 4)$. DATA RATE: 2.66 kb/s PER BAND, 16 kb/s TOTAL

Band	Transmitter	Symbol errors / 9600 (equalization only)	Information bit errors after decoding / 3200	SNR (dB)
1	1	1346	0	7.44
	2	679	0	8.23
	3	996	0	7.74
	4	921	0	8.33
2	1	1724	322, 0	7.09
	2	1552	223, 0	7.29
	3	351	0, 0	8.75
	4	1414	0, 0	8.31
3	1	1667	166, 0, 0	8.02
	2	1452	0, 0, 0	6.93
	3	1719	660, 452, 0	5.97
	4	1461	0, 0, 0	8.08
4	1	1735	293, 0, 0, 0	5.94
	2	1402	0, 0, 0, 0	7.55
	3	1089	0, 0, 0, 0	7.33
	4	2905	1192, 1062, 1166, 1129	5.66
5	1	2042	739, 752, 347, 0	6.89
	2	1551	13, 0, 0, 0	8.04
	3	2070	835, 714, 177, 0	6.54
	4	1979	714, 395, 0, 0	7.44
6	1	1465	0, 0	6.71
	2	757	0, 0	7.31
	3	2419	994, 1109	3.34
	4	1936	613, 0	7.125

Multiple numbers in any column refer to multiple iterative equalizer passes. Each transmitter uses a rate 1/3 turbo code, which results in 3200 information bits corresponding to 9600 coded bits. SNR is computed at the equalizer output.

the packet. When the SIC DFE is used, the detection order is determined by preprocessing the initial small block of training symbols and measuring the SNR at the equalizer output for each transmitter. The transmit streams are then processed in decreasing SNR order (highest SNR first, lowest SNR last). The training overhead incurred was 30%, although our decoding results have shown that 20% training gave similar performance at the cost of an increased number of turbo iterations (i.e., increased decoding latency).

1) *Decoding Results for SISO Transmissions:* We start by analyzing the results of a single transmitter BPSK data set with rate 1/2 turbo code and a channel interleaver. The estimated multipath-delay profile is shown in Fig. 13 which shows a principle arrival and a strong secondary arrival around the 20th symbol. The phase estimate obtained from the PLL is shown in Fig. 14. The error rate results are reported in Table I which shows that the bands could be decoded without errors. Moreover, although the equalizer output SNR reduces at higher frequencies due to frequency-dependent attenuation, the turbo decoder is still able to recover all the bits reliably. The reliability of the bit decisions can also be appreciated from Fig. 15 which shows the distribution of the information bit LLRs, as computed by the turbo

decoder. Moreover, the LLR distribution resembles the mixed Gaussian distribution very closely, indicating that ISI has been successfully mitigated leaving only the desired signal and noise at the equalizer output.

Before decoding each SISO packet, we selected the receiver parameters by preprocessing a few initial training symbols using the SIC algorithm. Table II shows the error rate result for all receivers and frequency bands for a single transmitter, turbo-coded BPSK data set using the previously chosen receiver settings. Once chosen, these settings were not changed manually as the decoder processed each receiver and each band one-by-one. We see that using this technique, most of the packets could be decoded with no errors (except a few in the higher bands due to very low SNR) indicating that the channel estimates obtained from the SIC algorithm can be used to efficiently initialize the receiver.

2) *Decoding Results For LSTC Transmissions:* In this section, we present the decoding results for the turbo-coded LSTC data sets. First, we consider the four-transmitter, LSTC data set with BPSK modulation. The decoding results obtained using MIMO DFE and SIC DFE are shown in Tables III and IV, respectively. These results are obtained by using four receivers op-

TABLE IV
DECODING RESULT FOR BPSK, TURBO-CODED, LSTC USING SIC DFE. DETECTION ORDER = $[2 \ 1 \ 3 \ 4]$. $(n_T, n_R) = (4, 4)$.
DATA RATE: 2.66 kb/s PER BAND, 16 kb/s TOTAL

Band	Transmitter	Symbol errors / 9600 (equalization only)	Information bit errors after decoding / 3200	SNR (dB)
1	1	522	0	7.18
	2	271	0	7.77
	3	1132	0	7.10
	4	893	0	7.42
2	1	1628	62, 0	6.49
	2	558	0, 0	6.98
	3	256	0, 0	7.92
	4	641	0, 0	7.65
3	1	701	0, 0	7.83
	2	1444	0, 0	6.22
	3	1738	532, 0	4.43
	4	1261	0, 0	7.02
4	1	2078	805, 682, 0, 0	6.45
	2	1593	2, 0, 0, 0	6.91
	3	728	0, 0, 0, 0	6.93
	4	1997	733, 562, 3, 0	6.17
5	1	1913	737, 4, 0	6.25
	2	840	0, 0	7.33
	3	1719	552, 0	5.89
	4	1624	206, 0	7.10
6	1	1111	0, 0	6.30
	2	691	0, 0	6.55
	3	2526	1111, 1061	2.87
	4	855	0, 0	6.52

Multiple numbers in any column refer to multiple iterative equalizer passes. Each transmitter uses a rate 1/3 turbo code, which results in 3200 information bits corresponding to 9600 coded bits. SNR is computed at the equalizer output.

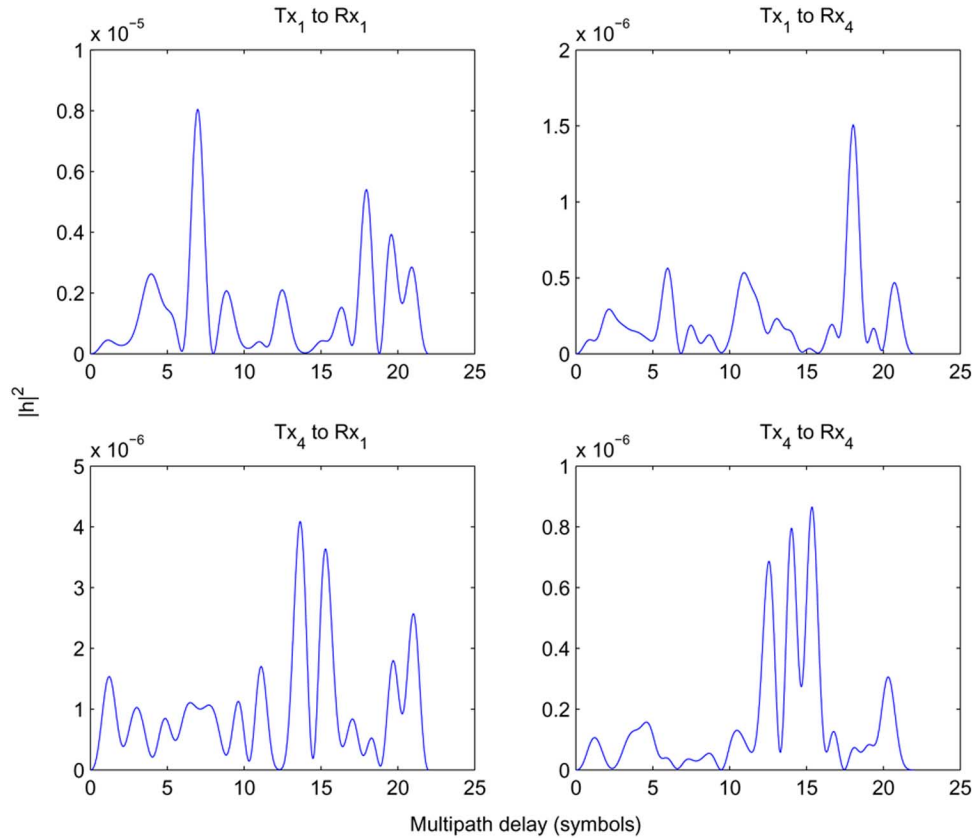


Fig. 16. Channel-impulse response estimates using the SIC algorithm for the shallowest and the deepest transmitters to the shallowest and the deepest receivers for BPSK, turbo-coded, four-transmitter LSTC data set. The transmitters and the receivers are indexed in an increasing order of depth, 1 being the shallowest.

timally chosen before packet equalization. The overall data rate for this case is 16 kb/s. We observe that the MIMO DFE could

decode most of the packets with zero errors except for transmitter 4 in band 4 and transmitter 3 in band 6 due to low SNR

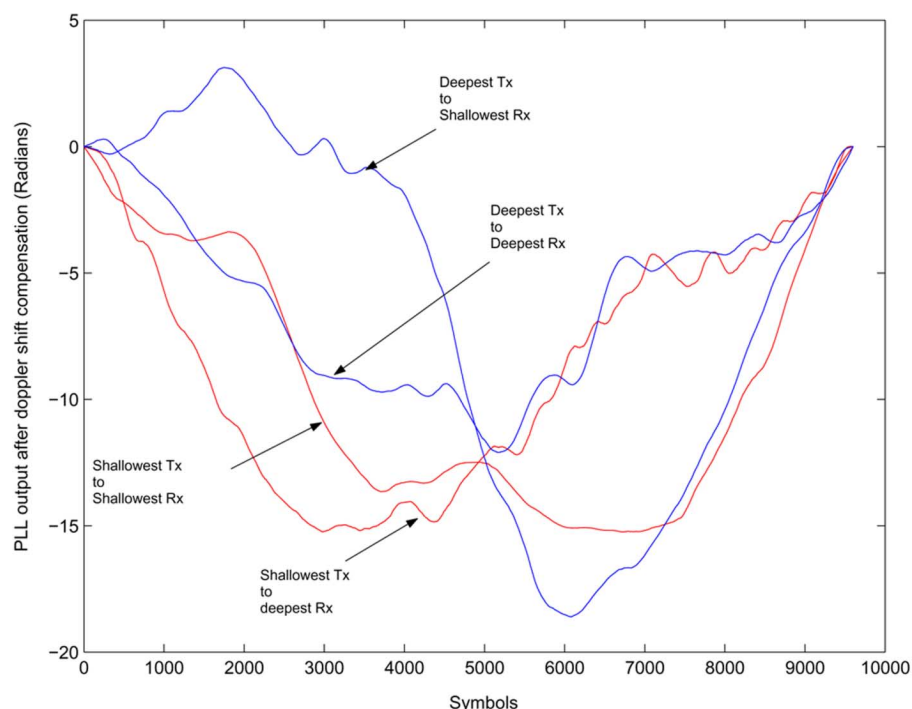


Fig. 17. Phase estimate at the PLL outputs for the shallowest and the deepest transmitters to the shallowest and the deepest receivers for BPSK, turbo-coded, four-transmitter LSTC data set.

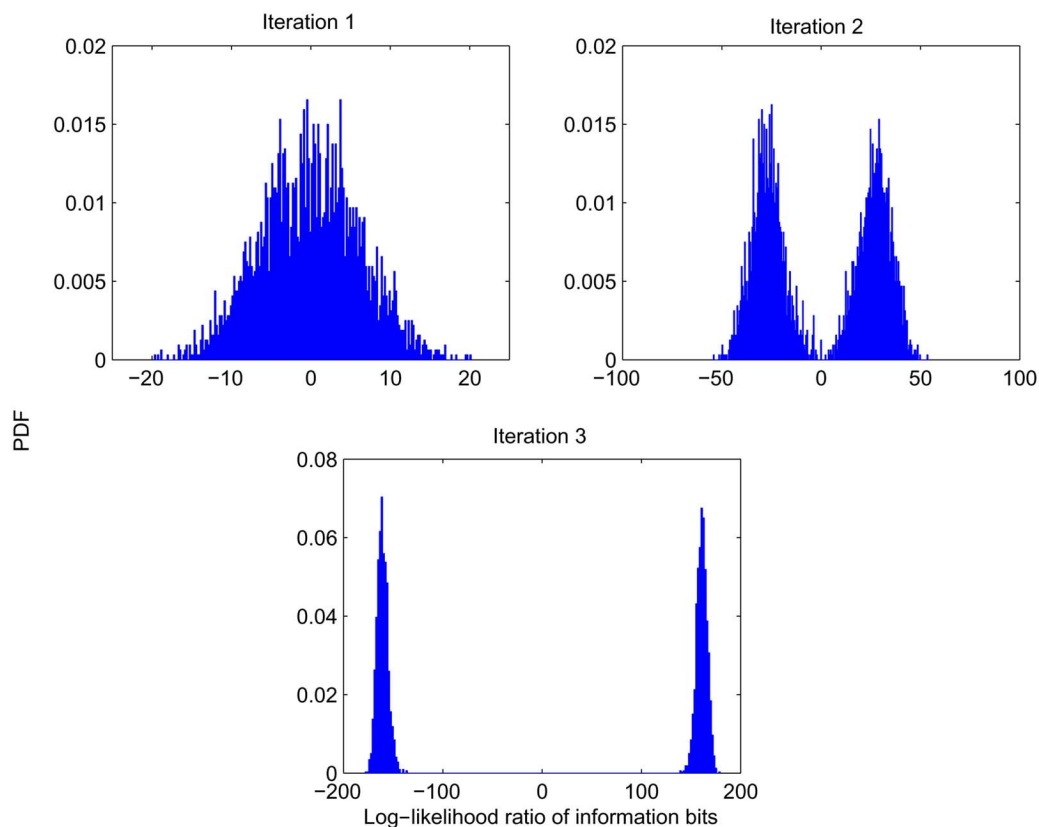


Fig. 18. Distribution of the information bit LLRs after 1, 2, and 3 iterations for BPSK, turbo-coded, four-transmitter LSTC data set.

at the receiver. However, the SIC DFE due to its shorter feedback section is less vulnerable to error propagation and hence is able to remove all the errors in transmitter 4 in band 4 after four turbo iterations. The channel estimates and the PLL outputs for the shallowest and the deepest transmitter–receiver pair

are shown in Figs. 16 and 17, respectively, with Fig. 16 showing that the deeper transmitters experience considerably more multipath than the shallower ones. Fig. 18 shows an example LLR distribution at the decoder output as a function of turbo iteration for band five. We observe that even with low operating

TABLE V
DECODING RESULT FOR QPSK, TURBO-CODED, LSTC USING SIC DFE. DETECTION ORDER = [1 2 4 3]. $(n_T, n_R) = (4, 6)$.
DATA RATE: 5.33 kb/s PER BAND, 32 kb/s TOTAL. TRANSMISSION 1, BLOCK 1

Band	Transmitter	Bit errors / 9600 (equalization only)	Bit errors after decoding / 3200	SNR (dB)
1	1	139	0	10.63
	2	22	0	11.67
	3	1159	0	7.59
	4	281	0	9.08
2	1	117	0	10.72
	2	154	0	10.11
	3	1367	0	6.2
	4	184	0	9.51
3	1	93	0	10.67
	2	69	0	10.17
	3	1607	14, 0	7.32
	4	260	0	9.30
4	1	868	0, 0, 0, 0, 0, 0, 0	9.10
	2	18	0, 0, 0, 0, 0, 0, 0	12.55
	3	2355	1059, 994, 879, 836, 641, 77, 0	6.01
	4	1713	0, 0, 0, 0, 0, 0, 0	8.22
5	1	2098	925, 868, 748, 407, 0	3.67
	2	196	0, 0, 0, 0, 0	9.98
	3	1151	0, 0, 0, 0, 0	8.13
	4	734	0, 0, 0, 0, 0	8.69
6	1	2078	879, 767, 680, 315, 0	8.36
	2	1072	0, 0, 0, 0, 0	8.01
	3	2531	1103, 1022, 1073, 1083, 1145	2.57
	4	1700	622, 0, 0, 0, 0	8.21

Multiple numbers in any column refer to multiple iterative equalizer passes. Each transmitter uses a rate 1/3 turbo code, which results in 3200 information bits corresponding to 9600 coded bits. SNR is computed at the equalizer output.

TABLE VI
DECODING RESULT FOR QPSK, TURBO-CODED, LSTC USING SIC DFE. DETECTION ORDER = [1 2 4 3]. $(n_T, n_R) = (4, 6)$.
DATA RATE: 5.33 kb/s PER BAND, 32 kb/s TOTAL. TRANSMISSION 1, BLOCK 2

Band	Transmitter	Bit errors / 9600 (equalization only)	Bit errors after decoding / 3200	SNR (dB)
1	1	23	0	11.52
	2	52	0	10.73
	3	1567	0	7.12
	4	389	0	8.47
2	1	153	0, 0, 0, 0	9.68
	2	94	0, 0, 0, 0	10.10
	3	2084	832, 740, 405, 0	3.44
	4	70	0, 0, 0, 0	10.33
3	1	106	0	9.93
	2	52	0	10.83
	3	961	0	7.22
	4	169	0	9.54
4	1	1416	7, 0, 0	8.82
	2	30	0, 0, 0, 0	12.4
	3	2208	942, 606, 0	3.3
	4	1315	0, 0, 0	9.32
5	1	3005	1215, 1180, 1183	2.42
	2	958	0, 0, 0	9.73
	3	1762	586, 0, 0	8.34
	4	487	0, 0, 0	9.84
6	1	2757	1107, 1064	2.78
	2	1667	548, 0	2.87
	3	3063	1209, 1170	2.51
	4	3270	1322, 1174	2.39

Multiple numbers in any column refer to multiple iterative equalizer passes. Each transmitter uses a rate 1/3 turbo code, which results in 3200 information bits corresponding to 9600 coded bits. SNR is computed at the equalizer output.

SNR, the quality of the LLRs becomes progressively better with each turbo iteration and eventually the distribution approaches a mixed Gaussian distribution.

We now focus on the four-transmitter LSTC data set with QPSK modulation. This gives us an overall rate of 32 kb/s with

5.33 kb/s per band. For this data set, two codewords were transmitted in each packet of each band for each transmitter. The decoding results for the two codewords are provided separately in Tables V–VIII. We observe that a large majority of the codewords are decoded with zero errors. However, for the higher bands, there

TABLE VII
DECODING RESULT FOR QPSK, TURBO-CODED, LSTC USING SIC DFE. DETECTION ORDER = [1 2 3 4]. $(n_T, n_R) = (4, 6)$.
DATA RATE: 5.33 kb/s PER BAND, 32 kb/s TOTAL. TRANSMISSION 2, BLOCK 1

Band	Transmitter	Bit errors / 9600 (equalization only)	Bit errors after decoding / 3200	SNR (dB)
1	1	646	0	9.5
	2	62	0	11.2
	3	125	0	10.2
	4	224	0	9.6
2	1	257	0	9.22
	2	40	0	11.34
	3	1561	0	8.21
	4	1049	0	7.77
3	1	244	0	9.47
	2	50	0	10.9
	3	399	0	9.79
	4	1298	0	8.44
4	1	1636	423, 0, 0	8.3
	2	119	0, 0, 0	10.6
	3	100	0, 0, 0	11.18
	4	1724	609, 427, 0	9.32
5	1	2587	1153, 1102, 1045	3.06
	2	113	0, 0, 0	10.42
	3	1846	775, 573, 0	3.16
	4	1455	0, 0, 0	7.81

Multiple numbers in any column refer to multiple iterative equalizer passes. Each transmitter uses a rate 1/3 turbo code, which results in 3200 information bits corresponding to 9600 coded bits. SNR is computed at the equalizer output.

TABLE VIII
DECODING RESULT FOR QPSK, TURBO-CODED, LSTC USING SIC DFE. $(n_T, n_R) = (4, 6)$. DATA RATE: 5.33 kb/s PER BAND, 32 kb/s TOTAL.
TRANSMISSION 2, BLOCK 2

Band	Transmitter	Bit errors / 9600 (equalization only)	Bit errors after decoding / 3200	SNR (dB)
1	1	726	0	8.86
	2	27	0	11.96
	3	1234	0	10.20
	4	1953	804, 395, 0	3.73
2	1	676	0	8.55
	2	109	0	10.57
	3	2264	934, 887, 832, 837, 795	3.23
	4	2076	815, 681, 595, 272, 0	5.65
3	1	759	0	8.85
	2	76	0	10.97
	3	1934	795, 600, 394, 0	4.07
	4	1318	0, 0, 0, 0	8.46
4	1	1800	745, 543, 407, 58, 0	8.32
	2	84	0, 0, 0, 0, 0	10.78
	3	1528	193, 0, 0, 0, 0	9.03
	4	3291	1240, 1247, 1158, 1199, 1141	2.63
5	1	1961	864, 768, 707, 597, 264, 0	3.00
	2	72	0, 0, 0, 0, 0	10.08
	3	1771	576, 0, 0, 0, 0	7.05
	4	2580	1081, 978, 1018, 1066, 1124, 1110	2.76

Multiple numbers in any column refer to multiple iterative equalizer passes. Each transmitter uses a rate 1/3 turbo code, which results in 3200 information bits corresponding to 9600 coded bits. SNR is computed at the equalizer output.

are a few instances of decoder failure where the turbo equalizer failed to converge. This is in sharp contrast with STTC where spatial encoding ensures full transmit diversity. In that case, since the two codeword components corresponding to the same information sequence are transmitted via two different transmitters, the overall information bit error probability converges to an acceptable value even if one of the transmitters observes a very low SNR. In LSTC, however, there is no encoding across transmitters and sustained deep fades for one transmitter can cause it to fail completely as is seen for several of the high band transmissions of this data set. This is the classical diversity versus multiplexing

gain tradeoff that we have also observed in our experiments. The channel impulse response estimate and the PLL output for this case are shown in Figs. 19 and 20. The distribution of the information bit LLRs is shown in Fig. 21. Comparing it with the LLR distribution of nonturbo-coded STTC, we observe that the average magnitude of the LLRs tend to be large for turbo-coded LSTC. This is because the turbo decoder is itself an iterative structure and, with each iteration within the turbo decoder, the LLRs become more reliable (i.e., increases in magnitude).

3) *Decoding Results For STTC Transmissions:* Now, we illustrate the performance of STTC data sets using QPSK

TABLE IX
DECODING RESULT FOR QPSK, STTC USING MIMO DFE. $(n_T, n_R) = (2, 3)$. DATA RATE: 4 kb/s PER BAND, 24 kb/s TOTAL

Band	Transmitter	Symbol errors / 9600 (equalization only)	SNR (dB)	Group	Information bit errors after decoding / 19200
1	1	58	11.74	1	96, 18, 11
	2	1592	8.67		
2	1	237	10.25	1	416, 61 39
	2	2245	8.16		
3	1	111	11.43	1	220, 34, 21 17
	2	2808	8.23		
4	1	107	11.09	1	195, 30, 26 21
	2	2269	8.61		
5	1	2711	8.03	1	2587, 327, 205, 167
	2	2698	7.65		
6	1	4869	7.27	1	4948, 1783, 546, 356
	2	4018	6.52		

Multiple numbers in any column refer to multiple iterative equalizer passes. Each group transmitted 9600 QPSK symbols which corresponds to 19200 information bits. SNR is computed at the equalizer output.

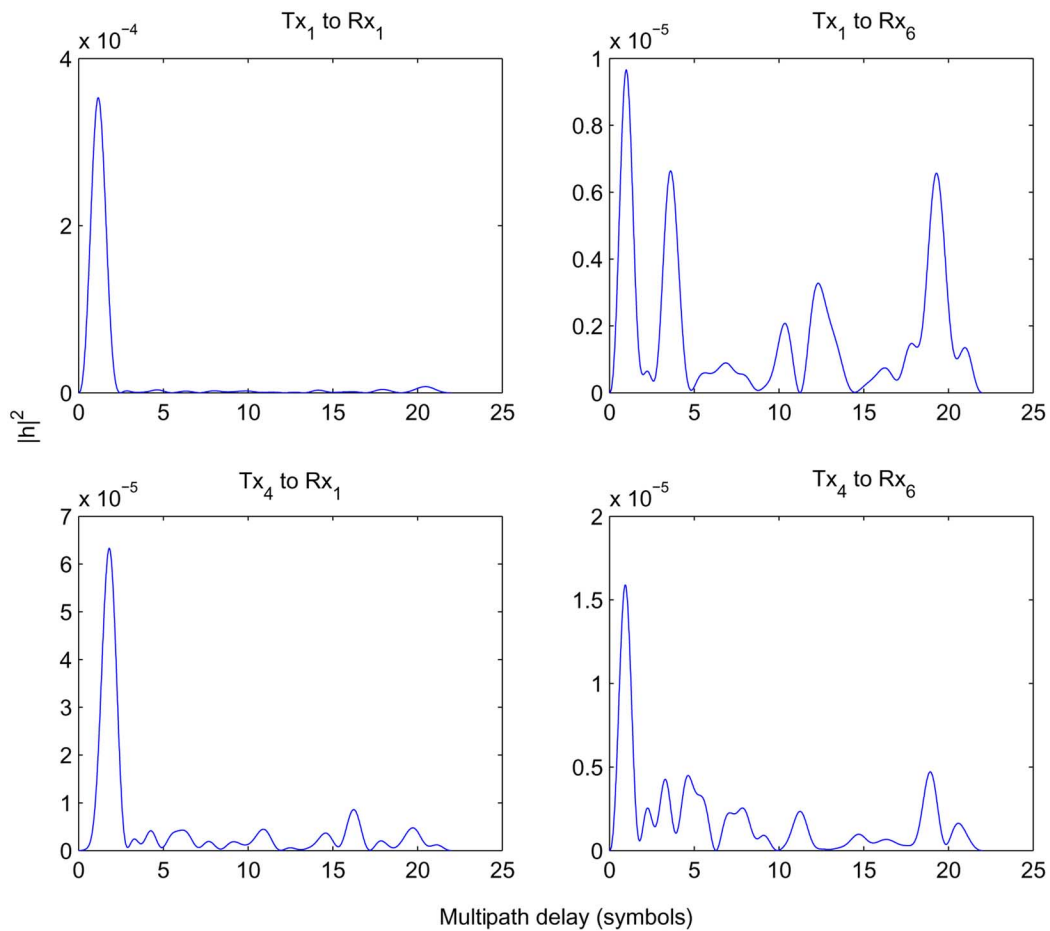


Fig. 19. Channel-impulse response estimates using the SIC algorithm for the shallowest and the deepest transmitters to the shallowest and the deepest receivers for QPSK, turbo-coded, four-transmitter LSTC data set. The transmitters and the receivers are indexed in an increasing order of depth, 1 being the shallowest.

modulation. First, we present the decoding results for a two-transmitter STTC case. In the grouped space-time trellis coding context, this corresponds to a single-group scheme. Table IX lists the error rates along with the equalizer output SNRs for the six bands. These results are obtained with three receivers. Comparing the raw symbol error rates before decoding (third column) with the information BERs (sixth column), we immediately realize the benefits of transmit diversity and iterative equalization. For example, for band 1, the raw symbol error rate for transmitter 2 is more than an order of magnitude higher

than that of transmitter 1. However since STTC encodes the information across the transmitters (i.e., transmit diversity), the overall BER after decoding is less than 1% and reduces further with turbo iterations. Similarly, in band 6, both transmitters experience low SNR, due to which the error rate after the first equalizer iteration is high. However, this error rate reduces dramatically (by an order of magnitude) after four iterations.

We emphasize here that the space-time trellis-coded data sets did not have any outer code. In an operational system, it is relatively easy to reduce these error rates to an arbitrarily small

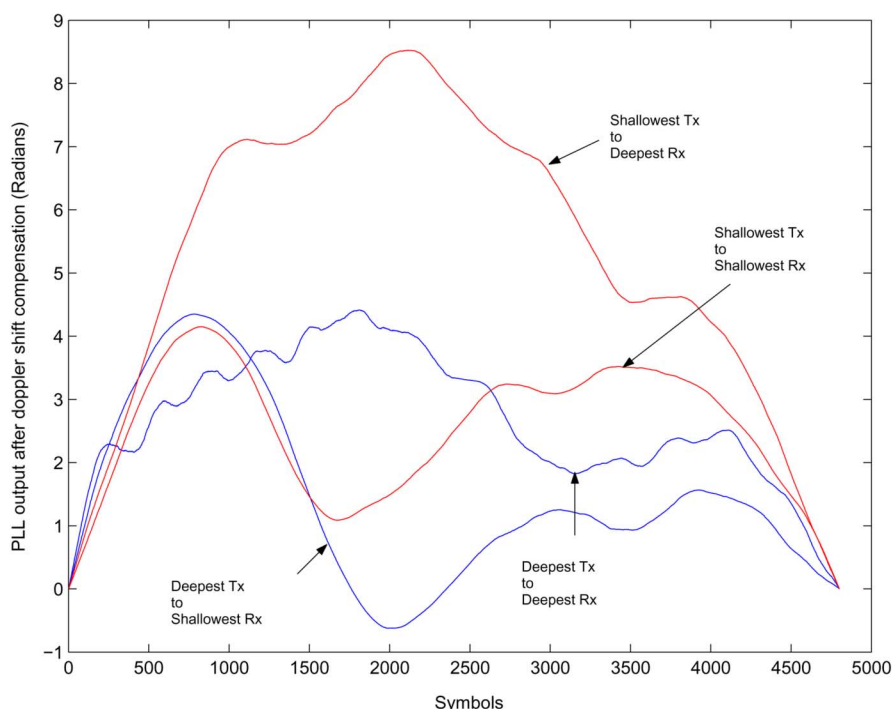


Fig. 20. Phase estimate at the PLL outputs for the shallowest and the deepest transmitters to the shallowest and the deepest receivers for QPSK, turbo-coded, four-transmitter LSTC data set.

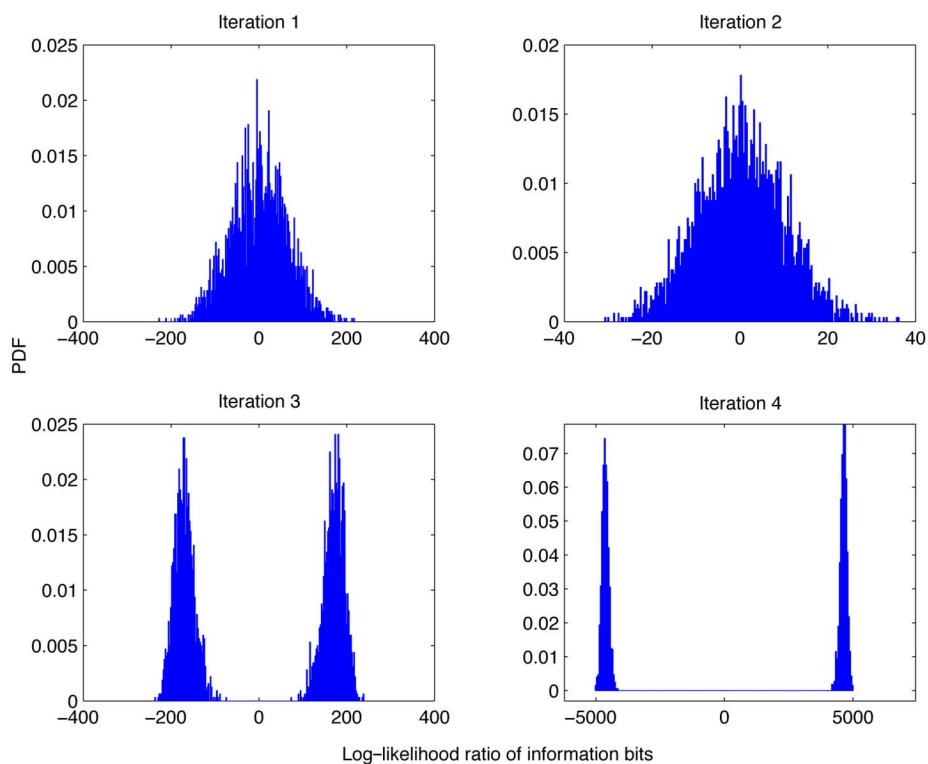


Fig. 21. Distribution of the LLRs of input bits after iteration 1, 2, 3, and 4 for QPSK, turbo-coded, four-transmitter LSTC data set.

value by preceding the STTC with a high-rate turbo code followed by an interleaver. As a specific example, consider the decoding result of band 5 in Table IX. The BER (without any outer code) for this case is $p = 167/19\,200 = 0.0087$. If we approximate the overall combination of the STTC, the UWA

channel, and the MIMO DFE as a binary symmetric channel (BSC) with crossover probability p , then the capacity of such a channel is given by $C = 1 + p \log(p) + (1-p) \log(1-p)$. Substituting $p = 0.0087$, we obtain the capacity of such a channel as $C = 0.928$. This means that by choosing a capacity approaching

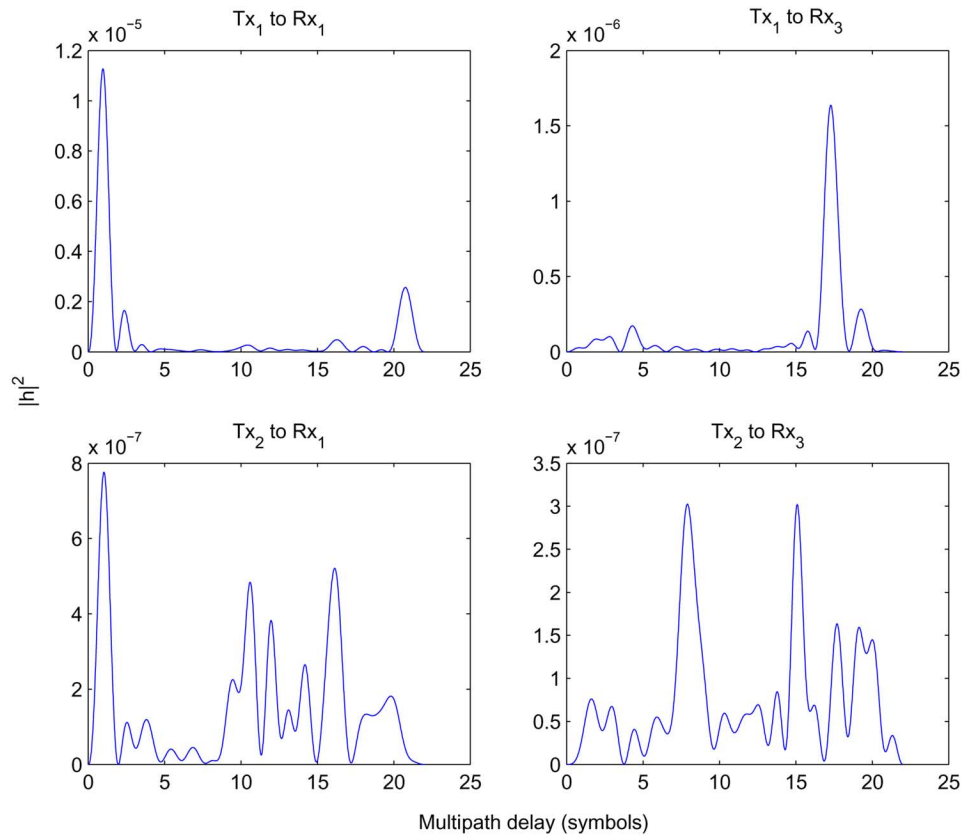


Fig. 22. Channel-impulse response estimates using the SIC algorithm for the shallowest and the deepest transmitters to the shallowest and the deepest receivers for two-transmitter, STTC QPSK data set. The transmitters and the receivers are indexed in an increasing order of depth, 1 being the shallowest.

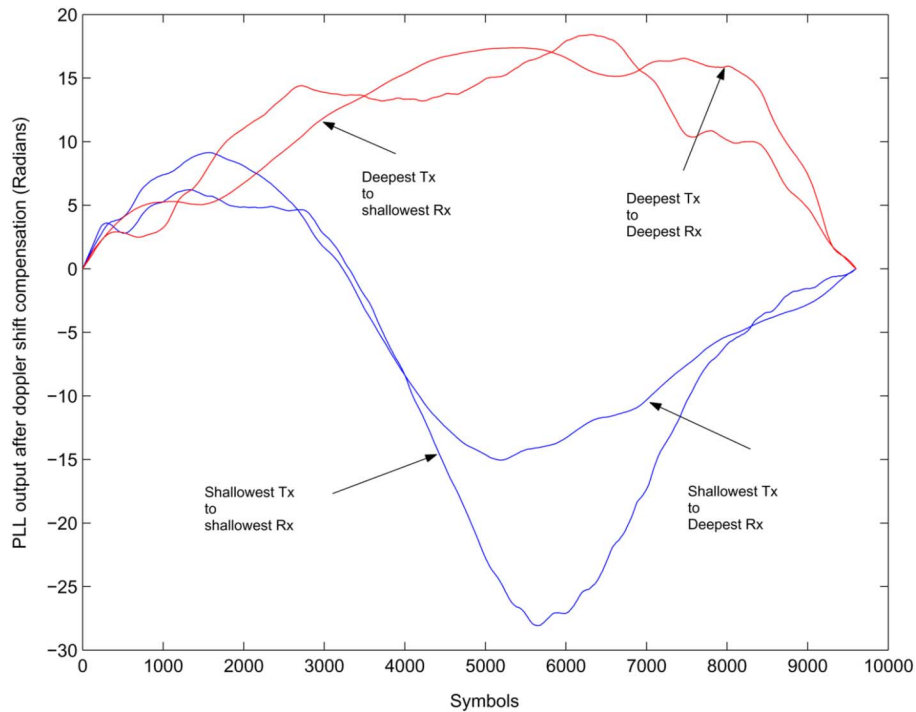


Fig. 23. Phase estimate at the PLL outputs for the shallowest and the deepest transmitters to the shallowest and the deepest receivers for two-transmitter, STTC QPSK data set.

code (such as a turbo code) with rate R slightly less than 0.928 as an outer code, we can reduce the error rates to an arbitrarily small value. Thus, if we had preceded the overall system with a turbo code with rate $R = 9/10$ followed by an interleaver of

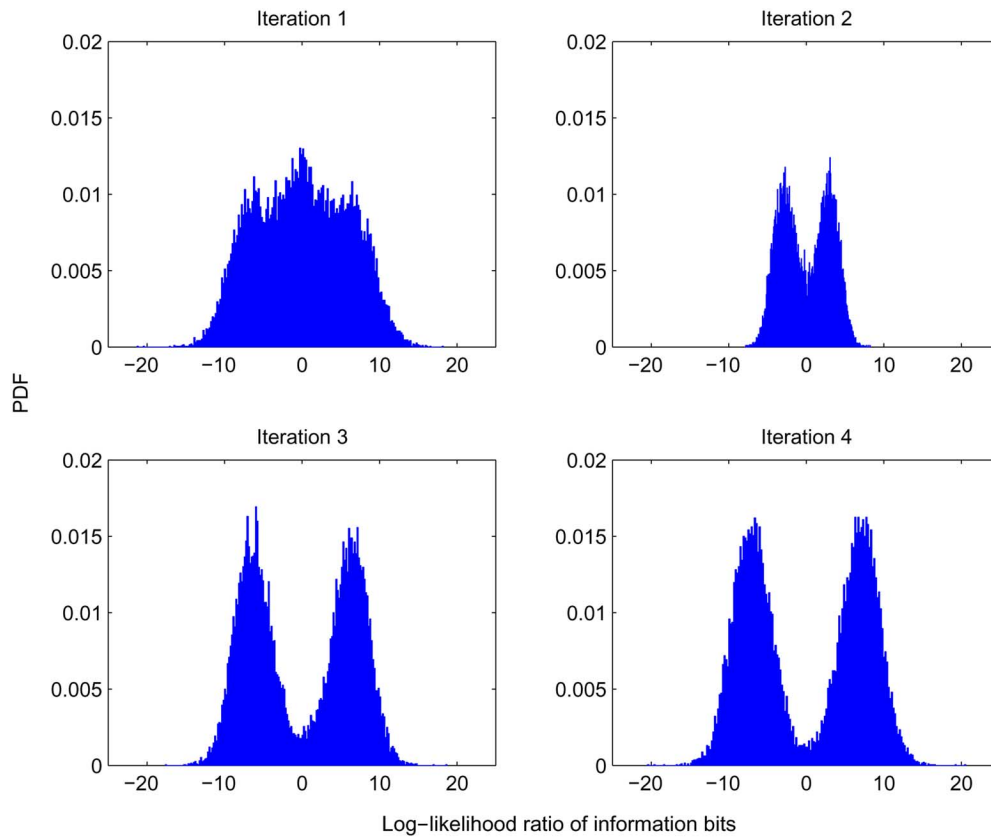


Fig. 24. Distribution of the LLRs of input bits after iteration 1, 2, 3, and 4 for two-transmitter, STTC QPSK data set.

TABLE X
DECODING RESULT FOR QPSK, STTC USING SIC DFE. $(n_T, n_R) = (4, 5)$. DATA RATE: 8 kb/s PER BAND, 48 kb/s TOTAL. TRANSMISSION 1

Band	Transmitter	Symbol errors / 9600 (equalization only)	SNR (dB)	Group	Information bit errors after decoding / 19200	Detection order
1	1	2509	8.31	1	3178, 468, 153, 133	[3 1 4 2]
	2	4438	6.78	2	1460, 117, 62, 58	
	3	490	9.22			
	4	3895	7.95		$P_e = 0.005$	
2	1	660	9.51	1	948, 1439, 197	[1 3 4 2]
	2	5288	4.20	2	1566, 942, 136	
	3	1351	8.99			
	4	4416	7.96		$P_e = 0.0087$	
3	1	187	10.33	1	281, 174, 87, 95, 87	[1 3 4 2]
	2	5149	4.32	2	4888, 2861, 756, 473, 391	
	3	4119	7.64			
	4	4961	5.75		$P_e = 0.012$	
4	1	1440	8.67	1	1884, 482, 195, 170, 135	[4 3 1 2]
	2	4470	5.93	2	3496, 652, 229, 175, 156	
	3	3460	8.38			
	4	3249	6.92		$P_e = 0.0076$	
5	1	3308	7.93	1	3501, 872, 301, 236	[3 4 1 2]
	2	3787	7.78	2	1730, 254, 97, 85	
	3	775	9.88			
	4	3741	8.52		$P_e = 0.0084$	
6	1	4546	7.23	1	5605, 3028, 1231, 789, 614	[3 4 1 2]
	2	4900	4.99	2	4037, 1998, 467, 365, 321	
	3	2887	8.25			
	4	4983	6.52		$P_e = 0.02$	

Multiple numbers in any column refer to multiple iterative equalizer passes. Each group transmitted 9600 QPSK symbols which corresponds to 19200 information bits. SNR is computed at the equalizer output. The probability of error P_e , shown for each band is computed by averaging the error probabilities of the groups in that band.

an arbitrary size, then the error rates would have gone to zero (with high probability) with only about 10% rate loss. To test this concept, we designed a turbo code of rate 9/10 such that its output codeword exactly matched the transmitted informa-

tion bit sequence. This can always be done by a random search method. Then, we passed the information bit LLRs produced by the iterative SIC DFE to the decoder of the newly designed turbo code. As expected, the error rate of $p = 0.0087$ at the output of

TABLE XI
DECODING RESULT FOR QPSK, STTC USING SIC DFE. $(n_T, n_R) = (4, 5)$. DATA RATE: 8 kb/s PER BAND, 48 kb/s TOTAL. TRANSMISSION 2

Band	Transmitter	Symbol errors / 9600 (equalization only)	SNR (dB)	Group	Information bit errors after decoding / 19200	Detection order
1	1	2255	9.22	1	1682 249 114 94 95	[4 1 2 3]
	2	2253	9.38	2	1747 123 96 46 35	
	3	4202	7.63			
	4	224	11.9		$P_e = 0.0033$	
2	1	1635	9.89	1	1485 246 101 81	[4 1 2 3]
	2	2987	8.39	2	3034 778 358 104	
	3	4834	7.22			
	4	392	11.00		$P_e = 0.0048$	
3	1	3796	9.43	1	4665 1518 364 176 134	[1 2 3 4]
	2	4683	8.37	2	4790 1933 795 324 224	
	3	4704	7.39			
	4	3717	8.55		$P_e = 0.0093$	
4	1	4728	7.32	1	3711 1297 386 324 259	[1 2 3 4]
	2	2883	8.74	2	5338 2373 794 262 160	
	3	4809	7.64			
	4	4530	9.33		$P_e = 0.01$	
5	1	5045	5.44	1	5337 2259 1079 744 691	[1 2 3 4]
	2	4432	8.01	2	5287 2664 1059 648 418	
	3	4985	6.60			
	4	4413	8.35		$P_e = 0.028$	
6	1	4641	4.05	1	5537 3302 2996 2121 1583	[1 2 3 4]
	2	4810	6.64	2	5828 2917 1724 911 980	
	3	4873	5.09			
	4	4893	7.54		$P_e = 0.066$	

Multiple numbers in any column refer to multiple iterative equalizer passes. Each group transmitted 9600 QPSK symbols which corresponds to 19200 information bits. SNR is computed at the equalizer output. The probability of error P_e , shown for each band is computed by averaging the error probabilities of the groups in that band.

TABLE XII
DECODING RESULT FOR QPSK, STTC USING SIC DFE. $(n_T, n_R) = (4, 6)$. DATA RATE: 8 kb/s PER BAND, 48 kb/s TOTAL. TRANSMISSION 3

Band	Transmitter	Symbol errors / 9600 (equalization only)	SNR (dB)	Group	Information bit errors after decoding / 19200	Detection order
1	1	599	9.61	1	803 221 79 80	[3 1 2 4]
	2	3621	7.48	2	707 109 58 55	
	3	152	10.8			
	4	4692	6.42		$P_e = 0.0035$	
2	1	4720	7.51	1	4718 1997 638 350 283	[3 1 2 4]
	2	3925	6.88	2	4202 2655 806 454 337	
	3	3127	7.97			
	4	5055	5.47		$P_e = 0.016$	
3	1	2359	8.97	1	2704 706 212 184 180	[3 1 2 4]
	2	3939	6.58	2	1634 478 213 179 171	
	3	612	9.46			
	4	5087	5.42		$P_e = 0.0091$	
4	1	3849	7.24	1	4225 1512 797 492 414	[1 3 2 4]
	2	4482	6.09	2	4626 2546 905 540 395	
	3	3580	8.32			
	4	5156	4.93		$P_e = 0.021$	
5	1	3798	10.1	1	4205 1511 624 314 309 242 263 228 249 219	[1 3 2 4]
	2	3970	8.5	2	5666 3717 2184 1456 1038 748 631 690 578	
	3	4427	9.56			
	4	5147	6.07		$P_e = 0.02$	

Multiple numbers in any column refer to multiple iterative equalizer passes. Each group transmitted 9600 QPSK symbols which corresponds to 19200 information bits. SNR is computed at the equalizer output. The probability of error P_e , shown for each band is computed by averaging the error probabilities of the groups in that band.

the iterative SIC DFE reduced to zero at the output of the turbo decoder with only 10% rate loss. The same argument can be applied to all the subsequent STTC results and can also be used as a useful tool for an operational system design.

Figs. 22 and 23 show the channel-impulse response and the phase estimates for band 6 (lowest operating SNR). Fig. 24 shows the information bit LLR distribution after iteration 1, 2, 3, and 4 clearly showing the progressive improvement obtained with each turbo iteration.

We now focus our attention on the four-transmitter, QPSK data set where two groups of STTCs are transmitted in parallel. This amounts to a data rate of 8 kb/s over each of the 3-kHz bands for an overall data rate of 48 kb/s over 23 kHz. Tables X–XII show the decoded BERs obtained using the SIC DFE algorithm for transmissions 1, 2, and 3, respectively, where each transmission consisted of 38 400 information bits per band (a total of $38\,400 \times 6 = 230\,400$ b for the whole packet). For this data set, transmitters 1 and 2 comprised group 1, and transmit-

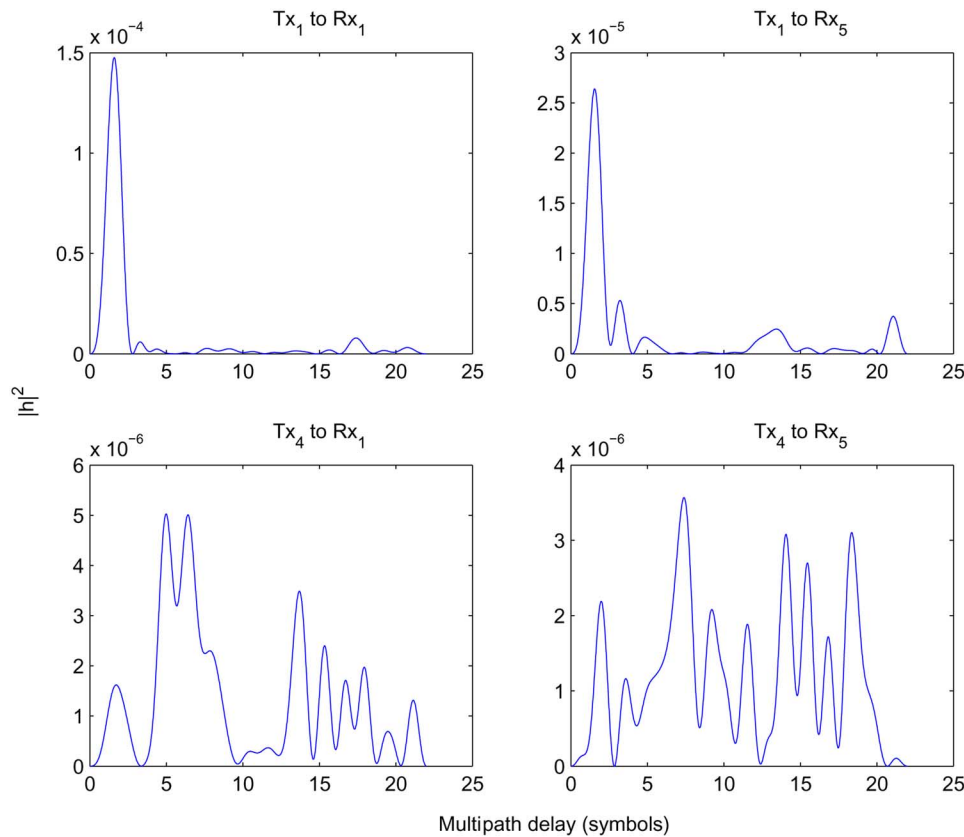


Fig. 25. Channel-impulse response estimates using the SIC algorithm for the shallowest and the deepest transmitters to the shallowest and the deepest receivers for four-transmitter, STTC QPSK data set. The transmitters and the receivers are indexed in an increasing order of depth, 1 being the shallowest.

ters 3 and 4 comprised group 2. We see from the tables that, in most cases, the iterative equalizer converges to 1% or less BER after three to four iterations which, as just mentioned, can be easily reduced to an arbitrarily small number by preceding the STTC with a high-rate turbo code and a large interleaver. Also, for some transmissions, the optimal detection orders turned out to be different for different frequency bands. The importance of transmit diversity is pronounced for this data set. For a given group, even if one of the transmitters experience much lower SNR than the other, the iterative equalizer was still able to reduce the overall BER for that group quickly. As an example, consider band 1 of transmission 3 (Table XII) where transmitters 1 and 2 form the first group (i.e., the STTC of group 1 is encoded across transmitters 1 and 2). We observe that even though transmitter 2 experiences close to a 40% symbol error rate after the first iteration, the overall BER for this group is eventually reduced to less than 0.5% due to high SNR experienced by transmitter 1.

Figs. 25 and 26 show the channel estimates and the PLL output for the shallowest and deepest transmitter to the shallowest and the deepest receivers. We observe that the multipath extent for the channel between the shallowest transmitter to the shallowest and the deepest receiver is less than that from the deepest transmitter. Fig. 27 shows the evolution of information bit LLRs as a function of turbo equalizer iterations. Similar behavior was observed for all the decoded data sets.

Next, we present the decoding results for a six-transmitter QPSK STTC data set in Table XIII. This data set provides a data

rate of 12 kb/s in each 3-kHz band yielding a spectral efficiency of $4 \text{ b} \cdot \text{s}^{-1} \cdot \text{Hz}^{-1}$. Over the entire band, this set would have provided a rate of 72 kb/s, but due to excessively low SNR and high cochannel interference the last two bands experienced unacceptably high error rates. Hence, the effective data rate for this set is 48 kb/s (4/6 of 72 kb/s). However, the acceptable error rate performance of the lower four bands suggest that a six-transmitter QPSK STTC scheme is a realistic means of obtaining high spectral efficiencies at lower frequencies for narrowband systems having only a few kilohertz of bandwidth.

Finally, we present the results for a two-transmitter, 8-PSK STTC data set in Table XIV. This data set provides a data rate of 6 kb/s in each of the 3-kHz band, giving a total data rate of 36 kb/s.

VIII. SUMMARY AND DISCUSSIONS

In this paper, we have demonstrated the usefulness of space-time coding combined with iterative equalization to achieve very high data rates over the bandlimited UWA communications channel. Specifically, we have investigated coding techniques including space-time trellis coding, turbo-coded layered space-time coding, and grouped space-time trellis coding at the transmitter. For space-time trellis-coded signals, we have demonstrated high-rate turbo codes as efficient outer codes to reduce residual decoder errors with minimal rate loss. We have also investigated multiband transmissions to obtain an effective compromise between equalization complexity

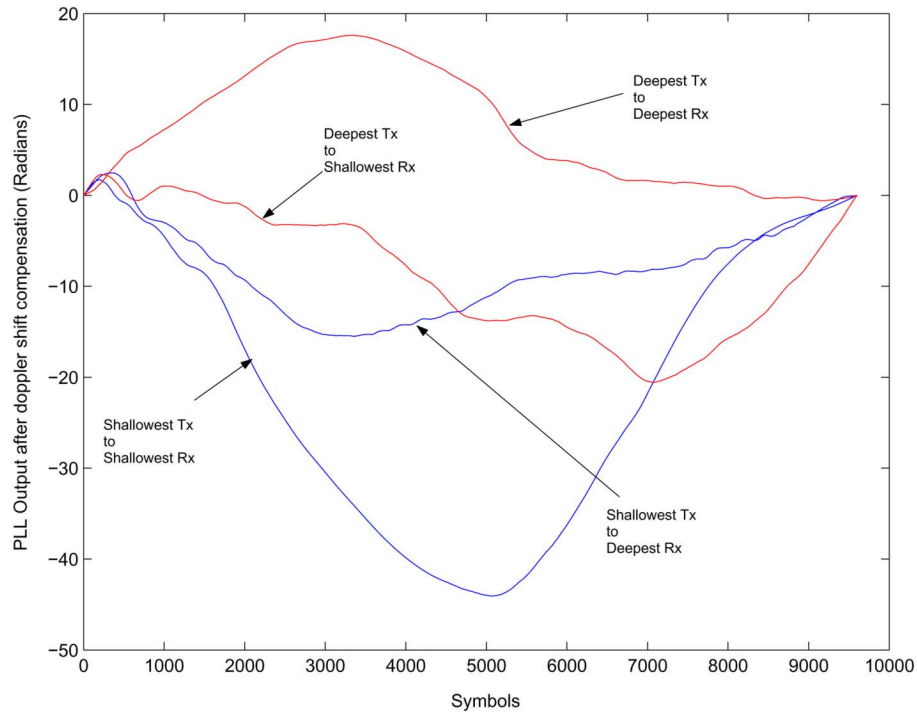


Fig. 26. Phase estimate at the PLL outputs for the shallowest and the deepest transmitters to the shallowest and the deepest receivers for four-transmitter, STTC QPSK data set.

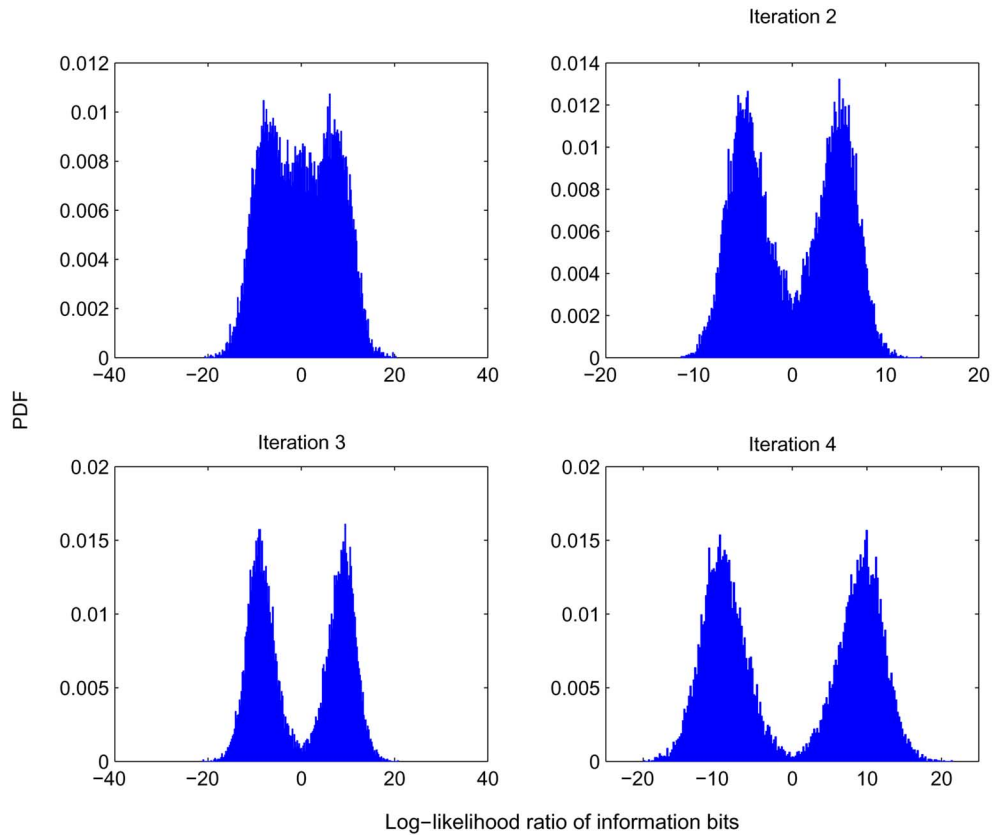


Fig. 27. Distribution of the LLRs of input bits after iteration 1, 2, 3, and 4 for four-transmitter, STTC QPSK data set.

and the tracking ability of the receiver. We have investigated iterative equalization as a powerful means to combat inter-symbol interference, cochannel, and rapid time variations in

the MIMO UWA channel. We have developed a MIMO DFE with an embedded Viterbi algorithm for STTC-encoded data packets. We have also developed an iterative MIMO DFE and

TABLE XIII
DECODING RESULT FOR QPSK, STTC USING SIC DFE. DETECTION ORDER = [1 5 2 6 3 4]. $(n_T, n_R) = (6, 7)$.
DATA RATE: 12 kb/s PER BAND ($4 \text{ b} \cdot \text{s}^{-1} \cdot \text{Hz}^{-1}$). TRANSMISSION 2

Band	Transmitter	Symbol errors / 9600 (equalization only)	SNR (dB)	Group	Information bit errors after decoding / 19200
1	1	656	10.23	1	916 396 83 81 73 74 70
	2	3888	7.01	2	4973 2542 1363 774 656 567 532
	3	4491	6.82	3	1820 404 162 132 142 142 142
	4	4590	6.09		
	5	1571	9.35		
	6	4031	7.3		$P_e = 0.012$
2	1	1034	10.08	1	1245 416 174 172 193 155 169
	2	3752	6.35	2	4309 1875 1003 636 593 614 539
	3	4049	6.3	3	3756 1204 445 318 302 277 284
	4	4223	7.0		
	5	3891	7.76		
	6	3645	7.76		$P_e = 0.017$
3	1	3179	10.86	1	3659 1515 373 233 183 194 175 178 168 174
	2	4173	7.7	2	5097 2743 1619 1195 881 838 802 819 784 747
	3	4839	5.47	3	4100 1374 383 233 202 197 205 196 198 197
	4	4367	6.8		
	5	3252	8.45		
	6	4536	7.4		$P_e = 0.019$
4	1	4353	7.58	1	5115 2391 1252 772 580
	2	4548	8.09	2	5258 2679 1365 1092 697
	3	4954	5.21	3	3338 921 263 208 176
	4	4348	7.6		
	5	2859	8.82		
	6	4300	6.87		$P_e = 0.025$

Multiple numbers in any column refer to multiple iterative equalizer passes. Each group transmitted 9600 QPSK symbols which corresponds to 19200 information bits. SNR is computed at the equalizer output. The probability of error P_e , shown for each band is computed by averaging the error probabilities of the groups in that band.

TABLE XIV
DECODING RESULT FOR 8PSK, STTC USING SIC DFE. $(n_T, n_R) = (2, 3)$ FOR BANDS 1,2,3,4 AND $(n_T, n_R) = (2, 4)$ FOR BANDS 5,6.
DATA RATE: 6 kb/s PER BAND, 36 kb/s TOTAL

Band	Transmitter	Symbol errors / 9600 (equalization only)	SNR (dB)	Group	Information bit errors after decoding / 28800
1	1	4294	10.22	1	4625, 1836, 54, 20
	2	4042	10.53		
2	1	4776	8.35	1	4330, 2700, 1383, 202, 110
	2	4142	9.13		
3	1	5411	8.56	1	6741, 4168, 1058, 197, 140, 129
	2	4825	8.57		
4	1	5686	8.86	1	6276, 4465, 983, 246, 83
	2	4017	9.47		
5	1	5586	10.82	1	7491, 5659, 2230, 112, 48
	2	4702	10.04		
6	1	5426	9.21	1	8720, 6664, 4513, 2022, 435, 240, 209
	2	6087	7.88		

Multiple numbers in any column refer to multiple iterative equalizer passes. Each group transmitted 9600 QPSK symbols which corresponds to 19200 information bits. SNR is computed at the equalizer output.

an iterative DFE with SIC for both LSTC- and STTC-encoded data packets. We have applied these algorithms to real data obtained from shallow-water, medium-range, UWA channels in the Pacific Ocean.

Our results from MakaiEx 2005 are summarized in Table XV and are compared with other historical achievements in data rates and spectral efficiencies in Table XVI (reproduced from [5]). In particular, the highest data rate that we have achieved is 48 kb/s over 23-kHz bandwidth over a 2-km link. This was achieved using a four-transmitter QPSK transmission using space-time trellis coding with no additional temporal coding. When an additional high-rate outer code (a rate 0.9 turbo code) is employed before space-time trellis coding, the residual bit errors of the STTC are reduced to zero at only 10% rate loss. We

also achieved 32 kb/s data rate using QPSK, four-transmitter, turbo-coded LSTC data with near-zero error rates. Finally, we have demonstrated a spectral efficiency of $4 \cdot \text{s}^{-1} \cdot \text{Hz}^{-1}$ using six-transmitter QPSK STTC transmission (12 kb/s over 3 kHz), which is useful for low-frequency UWA modems that are inherently narrowband. The achieved data rates and spectral efficiencies are significantly higher than other demonstrated systems at similar ranges and depths, as can be seen by comparing Tables XV and XVI, and hence, they demonstrate that multiple antenna transmissions along with sophisticated iterative decoding are capable of achieving data rates that are not possible with conventional single transmit antenna systems.

The data sets we have tested so far used fixed modulation and coding regardless of the channel conditions. However,

TABLE XV

SUMMARY OF THE MIMO DATA RATES AND SPECTRAL EFFICIENCIES OBTAINED FROM MAKAIEx 2005. SUBSCRIPT S INDICATES SHALLOW WATERS. NONZERO PROBABILITY OF ERROR FOR STTC IS REDUCED TO ZERO WHEN A RATE 0.9 TURBO CODE IS USED BEFORE SPACE-TIME CODING, RESULTING IN ONLY 10% RATE LOSS

Year	Data rate (kb/s)	Bandwidth (kHz)	Range (km)	Data type	Prob. of error
2005	6	23	2_S	1 tx, turbo coded BPSK	~ 0
2005	16	23	2_S	4 tx, turbo coded BPSK	~ 0
2005	32	23	2_S	4 tx, turbo coded QPSK	~ 0
2005	24	23	2_S	2 tx, STTC QPSK	$\sim 5 \times 10^{-3}$
2005	36	23	2_S	2 tx, STTC 8PSK	$\sim 3 \times 10^{-3}$
2005	48	23	2_S	4 tx, STTC QPSK	$\sim 10^{-2}$
2005	12	3	2_S	6 tx, STTC QPSK	$\sim 10^{-2}$

TABLE XVI

HISTORICAL DATA RATES AND IMPLIED SPECTRAL EFFICIENCIES DEMONSTRATED BY PHASE COHERENT SYSTEMS (REPRODUCED FROM [5]). N/A INDICATES NONAVAILABILITY OF DATA IN THE PUBLISHED REFERENCE. SUBSCRIPT S INDICATES SHALLOW WATERS AND SUBSCRIPT D INDICATES DEEP WATER, USUALLY A VERTICAL CHANNEL

Year	Data rate (kb/s)	Bandwidth (kHz)	Range (km)	Prob. of error
1989	500	125	0.06_D	$< 10^{-7}$
1985	20 – 30	10	3.5_D	$< 10^{-4}$
1993 – 94	0.6 – 3	0.3 – 1.0	$89 – 203_{S,D}$	$< 10^{-2}$
1994	6	3	0.04_S	N/A
1994	6	3	4.0_D	N/A
1994	20	20	0.09_S	$\sim 10^{-3}$
1995	1.1 – 2.2	0.6 – 2.2	$0.5 – 8.0_{S,D}$	$< 10^{-3}$
1996	0.2	0.2	50_D	$< 10^{-4}$
1997	0.9 – 1.8	N/A	$4.0_S – 8.0_D$	$< 10^{-4}$
1998	1.67 – 6.7	2 – 10	$4.0_S, 2.0_S$	N/A

the processing results clearly indicated that not all the MIMO links of the system experience similar average channel conditions. This is especially important for the layered space-time coded data sets where independent coding and modulation is employed for each transmitter. In such a scenario, if the transmitter could receive feedback from the receiver (either on an instantaneous basis, which is somewhat impractical, or on a statistical basis, which may be more practical), then the transmission parameters such as the code rate, the modulation scheme, the number of transmitters, and the transmit power level can be chosen adaptively. As an example, consider the four-transmitter QPSK LSTC scheme in Table V, where we see that transmitter 2 consistently observed much higher SNR than the rest of the transmitters across all the bands, with a higher SNR in excess of 10 dB in most cases. Clearly, a rate 1/3 code is highly redundant and unnecessary for this transmitter and a much higher rate code could have been used at the same transmit power level, or the transmit power level could have been reduced significantly at the same code rate to obtain similar performance. In the same way, for the grouped STTC data sets, some of the groups observed higher average SNR than the rest and could have provided similar performance with higher order modulation schemes. Thus, it is apparent that in a highly dynamic channel such as the UWA channel, adaptive modulation and coding added to MIMO signaling can improve the transmission rates even further. The combination of such adaptive techniques with MIMO signaling is a promising area of open research.

ACKNOWLEDGMENT

The Naval Research Laboratory provided the ACDS buoys used as the receive array.

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